The German University in Cairo
Electronics Dept., Faculty of IET
Course: Semiconductor (ELCT 503)
Assoc. Prof. Dr. Haitham Omran
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Semiconductors Project Report

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Part A:

Deck:

```
go atlas
# ~630nm GaInP/AlGaInP edge-emitter
# The Effect of High Compressive Strain on the Operation of
# AlGaInP Quantum-Well Lasers
# IEEE Journal of Quantum Electronics, VOL. 34, NO. 9, September 1998
# mesh space.mult=0.8 diag.flip

mesh  space.mult=1.0 diag.flip width=300 auto
#
x.mesh loc=0.00 spac=0.50
x.mesh loc=5.00 spac=0.50
#

region num=1 material=InAlGaP thick=1.5 sy=.15   x.comp=0.364 y.comp=0.156 \
         acceptor=5e18 bottom

region num=2 material=InAlGaP thick=0.095 sy=.01 x.comp=0.208 y.comp=0.312 \
         donor=2e14 bottom

region num=3 material=InGaP thick=0.01 sy=0.001  x.comp=0.516 \
         donor=2e14 bottom

region num=4 material=InAlGaP thick=0.095 sy=.01 x.comp=0.208 y.comp=0.312 \
         donor=2e14 bottom

region num=5 material=InAlGaP thick=1.5 sy=.15   x.comp=0.364 y.comp=0.156 \
         donor=1e18 bottom

elec num=1 name=anode top
elec num=2 name=cathode bottom

material region=1 eg300=2.35 NC300=9.13e+17 NV300=7.81e+18 \
         align=0.5 permittivity=11.39 epsinf=11.39 affinity=4.19 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10

material region=2 eg300=2.2 NC300=9.13e+17 NV300=7.81e+18 align=0.5 \
         permittivity=12.16 epsinf=12.16 affinity=4.26 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10

material region=3 taun0=1e-9 taup0=2e-8 copt=1.5e-10 epsinf=13.491 \
         align=0.5 eg300=1.92 affinity=4.4

material region=4 eg300=2.2 NC300=9.13e+17 NV300=7.81e+18 align=0.5 \
         permittivity=12.16 epsinf=12.16 affinity=4.26 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10

material region=5 eg300=2.35 NC300=9.13e+17 NV300=7.81e+18 align=0.5 \
         permittivity=11.39 epsinf=11.39 affinity=4.19 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10
```

```
models fldmob srh optr auger fermi consrh

models region=3 yan ^optr spontaneous qwell well.ny=120 well.cnbs=1 well.vnbs=1
# lorentz

mobility mun=1000 mup=500 vsatn=7.7e6 vsatp=7.7e6

solve init

solve previous

output band.param con.band val.band u.srh u.radiat u.auger flowlines

probe name="nwell" nwell x=2.5 y=1.6
probe name="pwell" pwell x=2.5 y=1.6

solve vstep=0.05 name=anode vfinal=1.8

#
# This case Photon Energy is extracted 1.93354 + 0.005 = 1.93854
#

models fldmob srh optr auger fermi consrh gainmod=5 ^print

laser trap maxtrap=10  proj rf=8 rr=90 maxch=100 \
      photon.energy=1.8 einit=1.5 efinal=2.0 lmodes \
      neff=3.2 cavity.length=300 sin=1e-5 index.mode=1 ^coupled

solve previous

log outf=laserex05.log

solve vstep=0.05 name=anode vfinal=1.9
log off

save outfile=laserex05_1.str

contact name=anode current

solve istep=0.5e-3  ifinal=1.5e-3 name=anode
log outf=laserex051.log
solve istep=0.5e-3  ifinal=5.5e-3 name=anode
log off

save outfile=laserex05_2.str
#
tonyplot laserex05.log
tonyplot laserex051.log

quit
```

1. Quantum Well Thickness

To find the thickness, we look at the region definitions in the code. The Quantum Well is the thin, narrow-bandgap layer sandwiched between the cladding layers.

- **Code Reference:**

region num=3 material=InGaP thick=0.01 sy=0.001 x.comp=0.516 \ donor=2e14 bottom

  - **Identification:** We know Region 3 is the Quantum Well because:
    1. It uses the material **InGaP** (as described in the project title).
    2. It is sandwiched between two **InAlGaP** layers (Regions 2 and 4).
    3. Later in the code, the command models region=3 ... qwell explicitly designates Region 3 as the quantum well.

- **Calculation:**
  - The parameter is thick=0.01.
  - In Silvaco ATLAS, dimensions are typically in **microns** (μm).
  - $0.01 \mu m = 10\text{nm}$.

Answer: The thickness of the InGaP Quantum Well is $\mathbf{0.01\ \mu m}$ (or **10 nm**).

2. Model and Bias Changes

1. Model Changes

In the previous assignment (Bulk Laser), we used standard models. In this QW code, specific commands are added to handle quantum confinement physics:

- **models ... qwell**: models region=3 ... qwell in the code. This activates the Schrödinger-Poisson solver to calculate the quantized energy levels (sub-bands) inside the well, rather than treating it as a bulk material.
- **gainmod=5**: In the laser models section, gainmod=5 is used. This is a specific gain model in Silvaco for quantum wells that accounts for the step-like density of states, whereas bulk lasers often use gainmod=1.

2. Bias Condition Changes

- **Code Reference:**

solve istep=0.5e-3 ifinal=5.5e-3 name=anode

This sweeps the current up to $\mathbf{5.5\ mA}$, which is much lower than the typical 100 mA range used for broad-area lasers in previous labs. This is because QW lasers have very low threshold currents due to the small active volume.

3. Energy Band Diagram & Bandgap Estimation

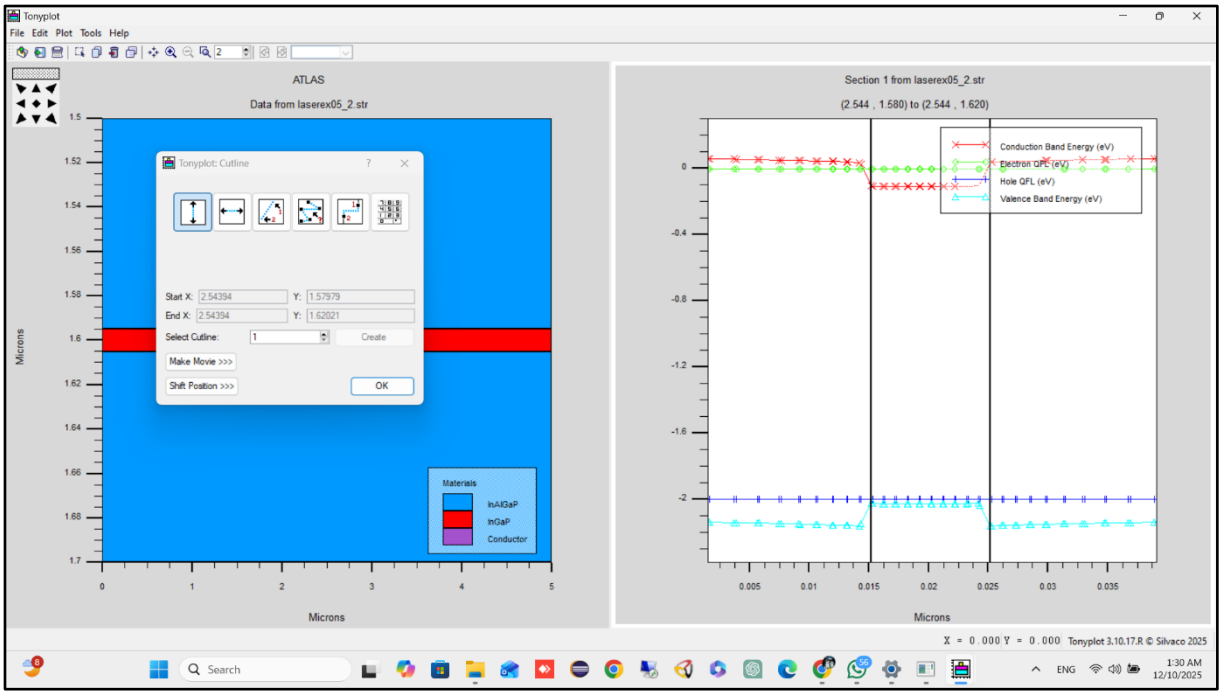


Figure [A1] InGaP Quantum Well Y-Cut Band-diagram

Observations:

- The band diagram zooms in on the thin InGaP Quantum Well (located centrally between the AlGaInP cladding layers).
- **Conduction Band (E_c, Red Line):** Creates a "well" shape, dropping to a minimum energy level of approximately $\mathbf{-0.12\ eV}$.
- **Valence Band (E_v, Cyan Line):** Rises to form a barrier, peaking at approximately $\mathbf{-2.05\ eV}$.

Estimation:

The energy gap (E_g) is the difference between the conduction and valence band edges inside the well:

$$E_g = E_c - E_v$$

$$E_g \approx (-0.12\ \text{eV}) - (-2.05\ \text{eV}) \approx 1.93eV$$

Verification:

This estimated value matches closely with the material parameter defined in the code for region 3 (eg300=1.92), which sets the bandgap of InGaP to 1.92 eV.

4. Checking for Population Inversion

Observation:

Examination of the band diagram reveals the following energy alignments:

- **Electron QFL (E_{fn}):** Located significantly **above** the Conduction Band edge (E_c), indicating strong electron degeneracy.
- **Hole QFL (E_{fp}):** Located very close to the Valence Band edge (E_v). Although it has not crossed deeply into the valence band, it remains at the band edge.

Verification of Inversion:

The fundamental requirement for stimulated emission is the condition:

$$E_{fn} - E_{fp} > E_g.$$

- From the graph: $E_{fn} \approx 0.0$ eV and $E_{fp} \approx -2.02$ eV, giving a separation of ≈ 2.02 eV.
- The bandgap is $E_g \approx 1.93$ eV.
- Since $2.02 \text{ eV} > 1.93 \text{ eV}$, the condition is satisfied, and population inversion is achieved. The strong degeneracy of the electron population compensates for the hole quasi-Fermi level being near the band edge.

5. Electron Concentration Comparison

Graphs:

- **Quantum Well:**

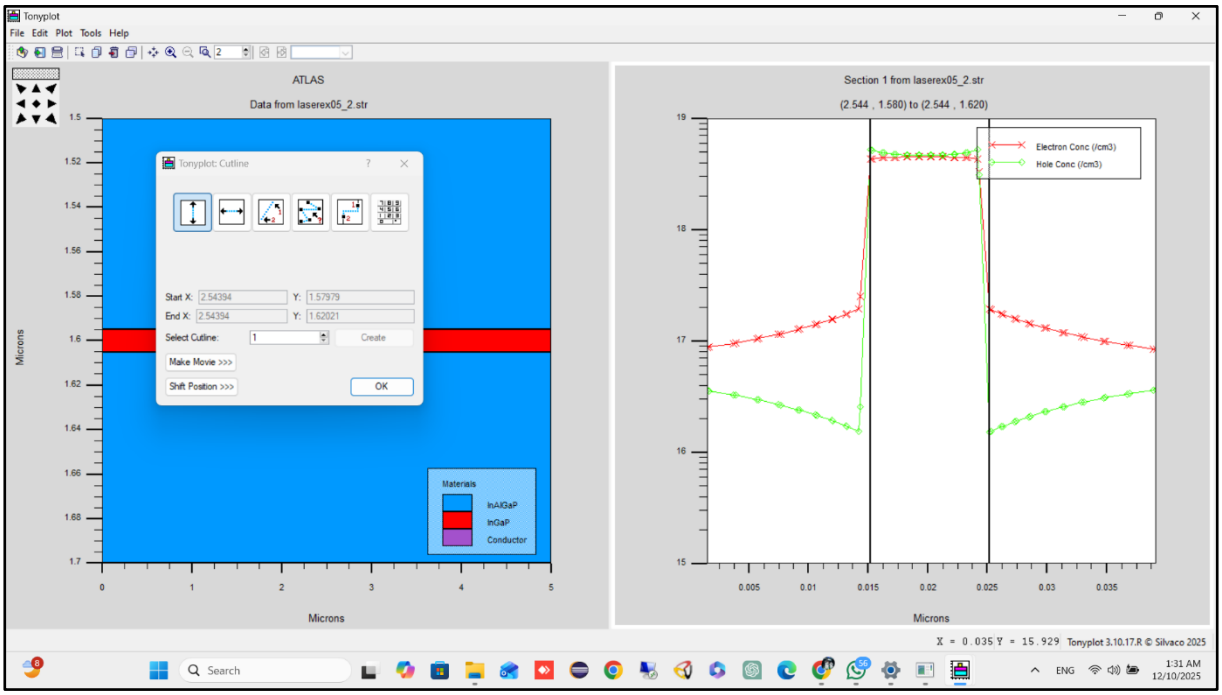


Figure [A2a] InGaP Quantum Well Carriers concentration

- **Bulk Laser (Assignment 2):**

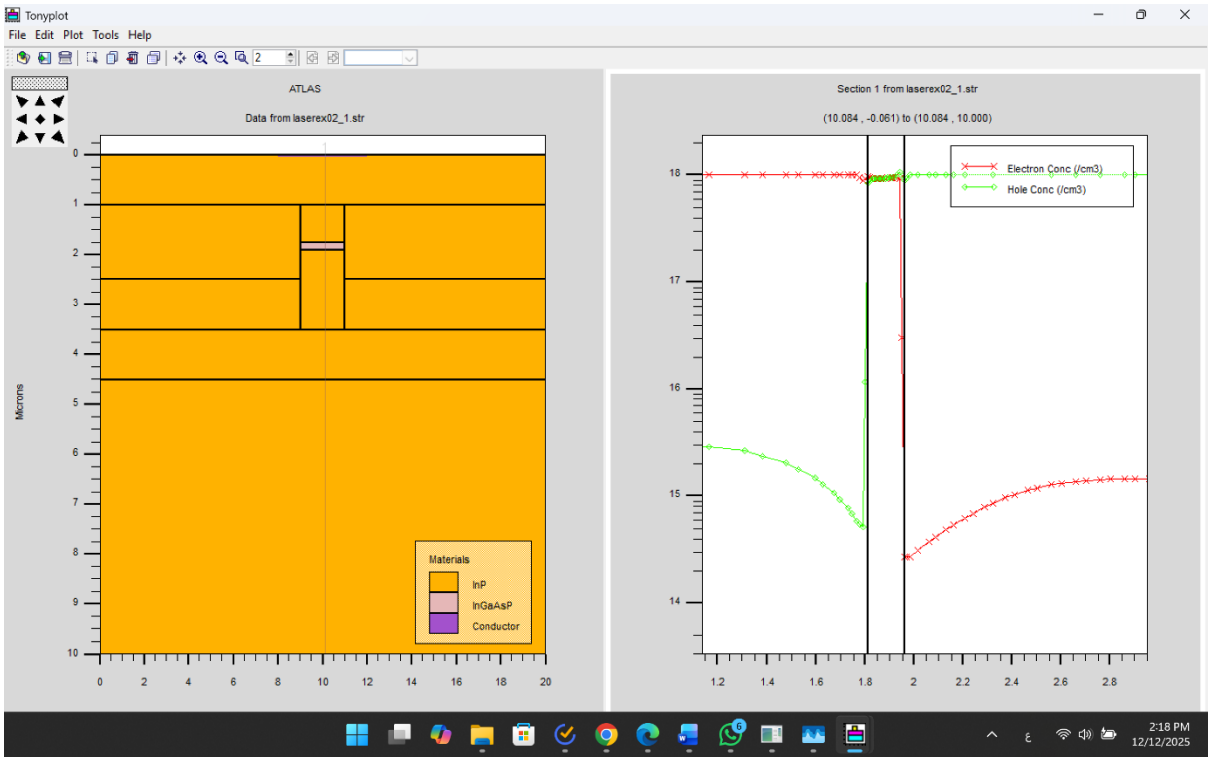


Figure [A2b] InGaAsP Buried Heterostructure Carriers concentration

Observation:

- **InGaP Quantum Well (Part A):** The electron concentration profile shows a sharp, delta-function-like spike confined strictly to the 10 nm well width. The peak concentration reaches a very high value (≈ 18.7 on log scale, or 5×10^{18} cm^{-3}) due to the extreme spatial confinement.
- **InGaAsP Bulk Laser (Assignment 2):** The electron concentration profile forms a broad, rectangular plateau across the thicker active region. The concentration is lower and distributed over a larger volume ($0.15\mu m$).

Comparison:

The Quantum Well structure provides superior carrier confinement. By trapping the carriers in a much narrower potential well (10 nm $VS \approx 150$ nm), the carrier density required for inversion is achieved more efficiently. The QW creates a localized high-density plasma that overlaps perfectly with the optical mode, whereas the bulk laser requires filling a larger volume to achieve the same gain density.

6. Recombination Rates Comparison

Graphs Used:

- **Quantum Well Laser (Part A):**

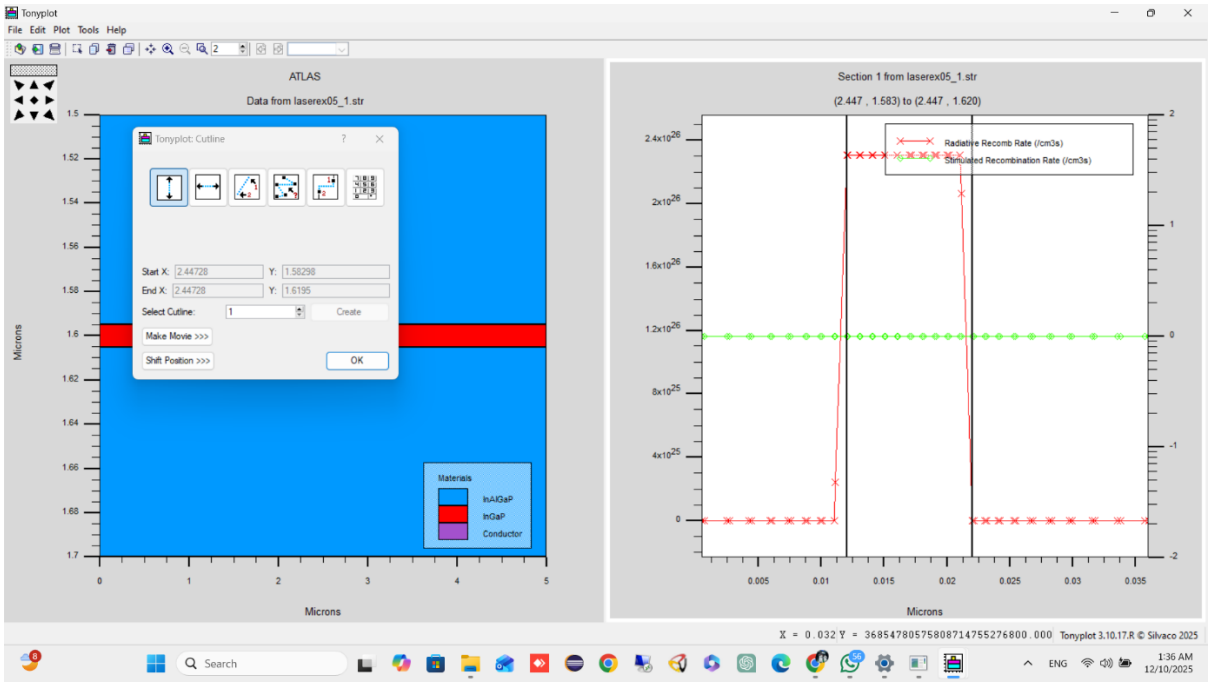


Figure [A3a]: InGaP Quantum Well Y-Cut Recombination Rates Before Lasing.

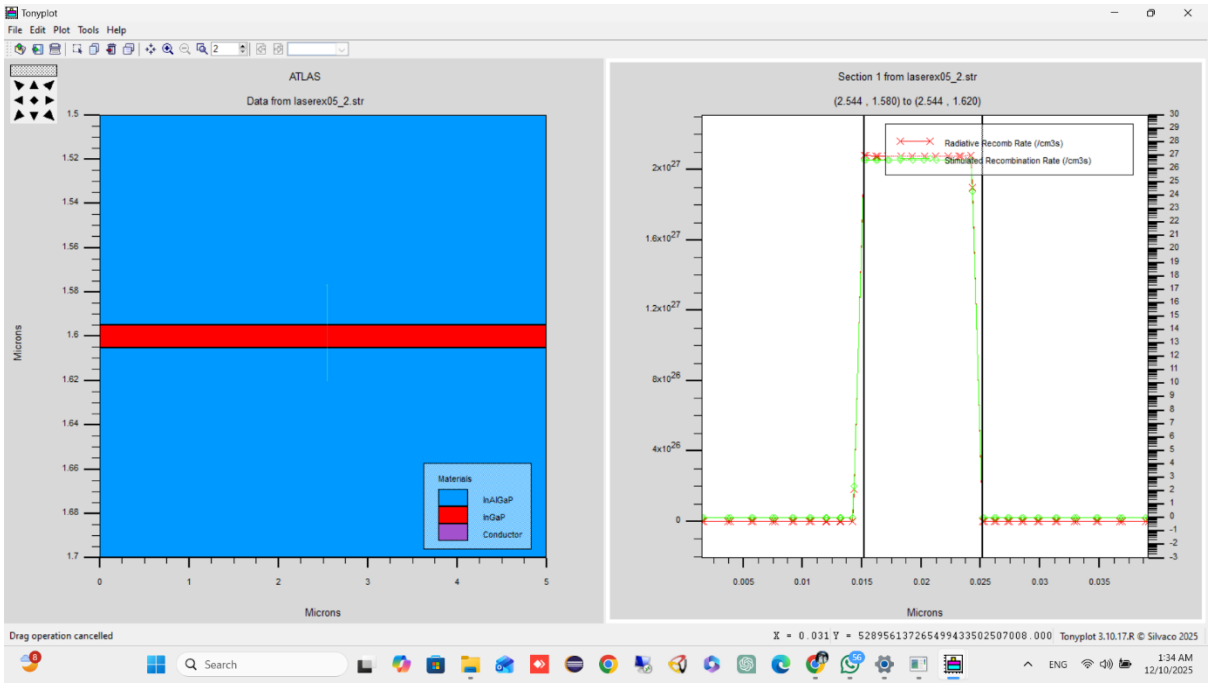


Figure [A3b]: InGaP Quantum Well Y-Cut Recombination Rates After Lasing.

- **Bulk Laser (Assignment 2):**

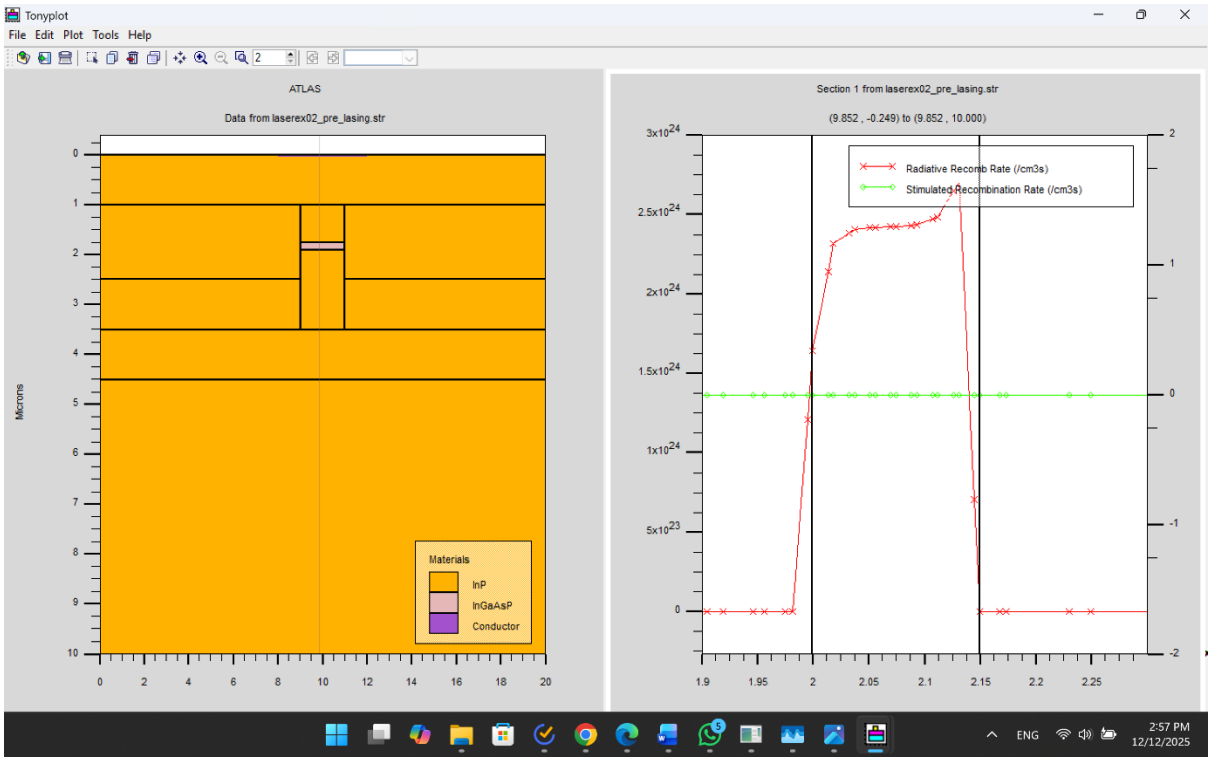


Figure [A3c]: InGaAsP Buried Heterostructure Y-Cut Recombination Rates Before Lasing.

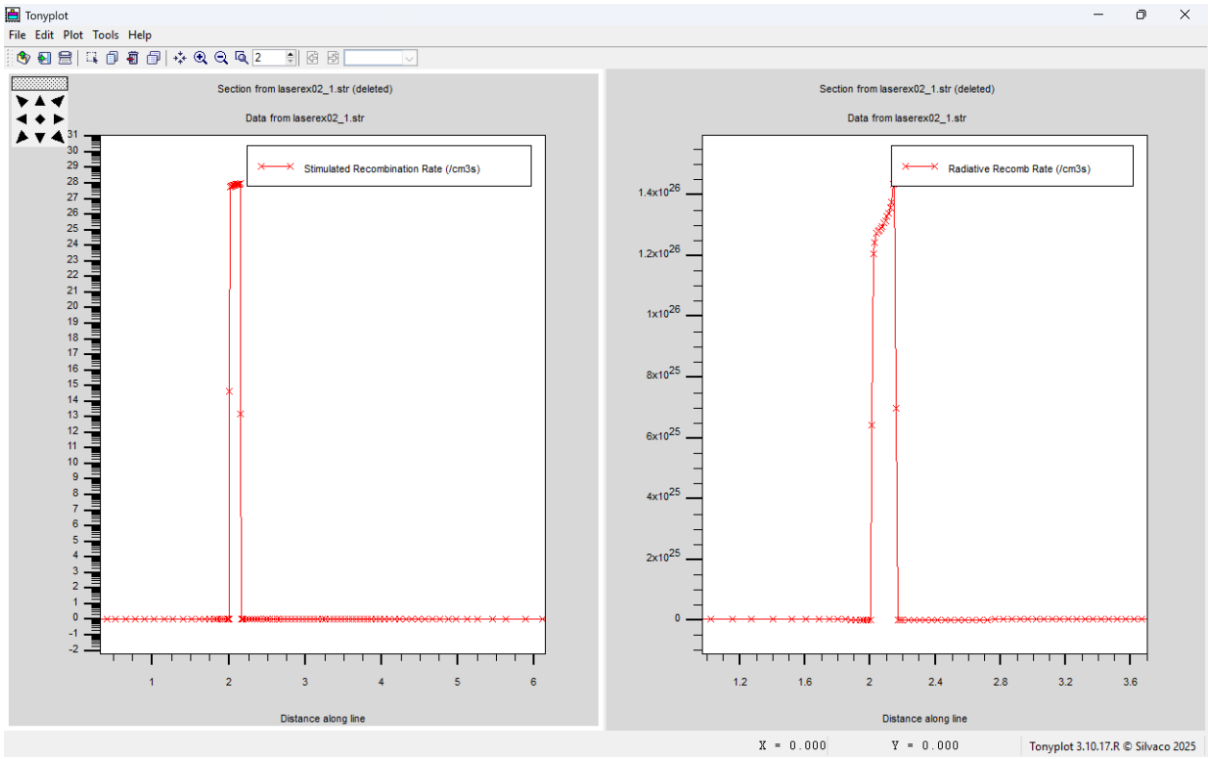


Figure [A3c]: InGaAsP Buried Heterostructure Y-Cut Recombination Rates After Lasing.

1. Before Lasing (Pre-Threshold)

Quantum Well (Part A):

- **Observation:**
  - **Radiative Recombination Rate (Red, Left Axis):** Confined to the ultra-narrow **10 nm** well, peaking at a massive $\approx 2.4 \times 10^{26}\ \mathrm{cm}^{-3s^{-1}}$.
  - **Stimulated Recombination Rate (Green, Right Axis):** The value is effectively **zero** (flatline on the log scale).
- **Comment:**
  - The device is operating purely in **Spontaneous Emission mode (LED mode)**.
  - Even though there is no lasing yet, the geometric confinement forces the radiative rate to be extremely high compared to bulk materials.

Bulk Laser (Assignment 2):

- **Observation:**
  - **Radiative Recombination Rate (Red, Left Axis):** Spreads across the broader **150 nm** active region, peaking at $\approx 2.5 \times 10^{24}\ \mathrm{cm}^{-3s^{-1}}$.
  - **Stimulated Recombination Rate (Green, Right Axis):** The value is effectively **zero**.
- **Comment:**
  - The carriers are diluted over a larger volume, resulting in a much lower local radiative rate.

Comparison:

At pre-lasing , the Quantum Well achieves a local radiative recombination rate that is 100 times higher ($10^{26}\ VS\ 10^{24}$) than the Bulk laser. This illustrates the massive boost in carrier interaction probability due to quantum confinement.

2. After Lasing (Post-Threshold)

Quantum Well (Part A):

- **Observation:**
  - **Radiative Rate:** Remains clamped near its threshold value ($\approx 2 \times 10^{27}$ $cm^{-3s^{-1}}$).
  - **Stimulated Recombination Rate (Green):** Spikes dramatically, matching or exceeding the radiative rate (typically reaching 10^{27} $cm^{-3s^{-1}}$ range).

Bulk Laser (Assignment 2):

- **Observation:**
  - **Stimulated Recombination Rate:** Rises to 10^{28} $cm^{-3s^{-1}}$ range.
  - The profile is a broad, rectangular block spanning the entire **150 nm** width.

- **Comment:**
  - **Stimulated Emission Dominance:** Once threshold is crossed, the stimulated rate shoots up from zero to become the dominant recombination mechanism. It consumes the majority of injected carriers, converting them into coherent photons.

7. Local Optical Gain Comparison

Graphs Used:

- **Quantum Well Laser (Part A):**

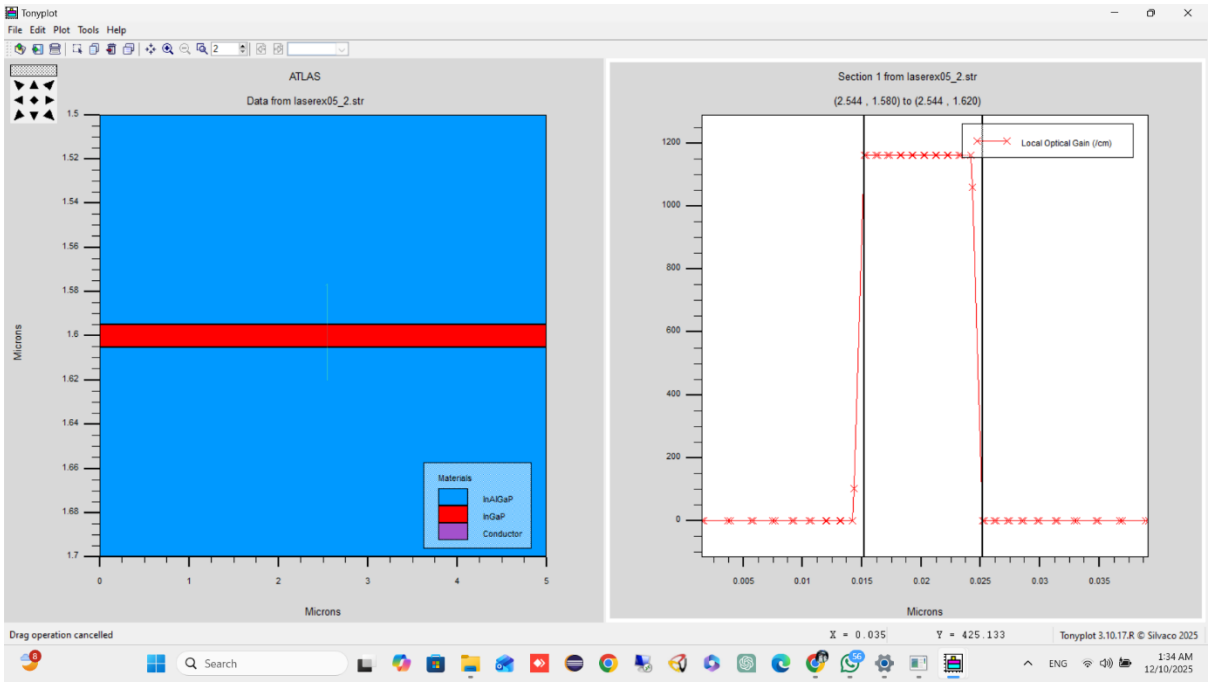


Figure [A4a]: InGaP Quantum Well Y-Cut Local Optical Gain.

- **Bulk Laser (Assignment 2):**

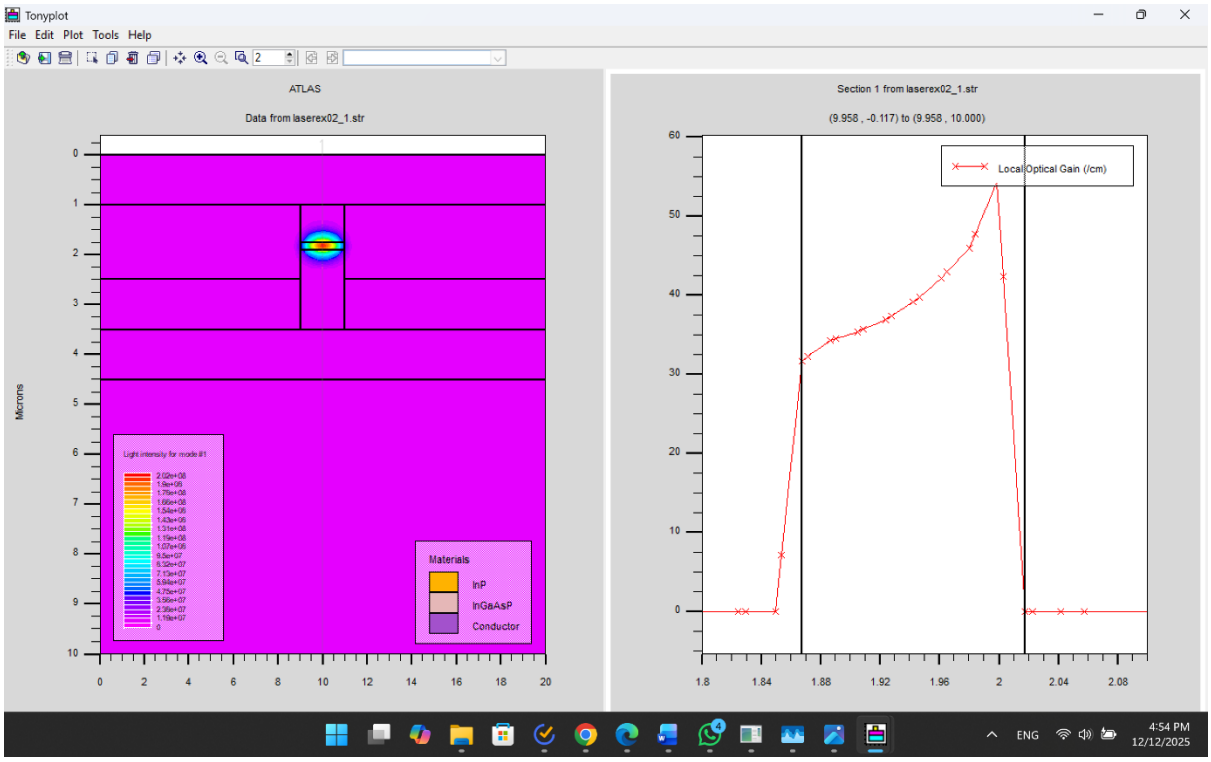


Figure [A4b]: InGaAsP Buried Heterostructure Y-Cut Local Optical Gain.

1. Observations

- **Quantum Well (Part A):**
  - **Peak Value:** The local optical gain reaches a massive value of approximately $\mathbf{1160\ cm^{-1}}$.
  - **Profile:** The gain profile is a perfect "box-like" shape, strictly confined to the narrow 10 nm well width $(0.015 - 0.025\mu m)$. The gain is uniform across the well.
- **Bulk Laser (Assignment 2):**
  - **Peak Value:** The local optical gain peaks at approximately $\mathbf{55\ cm^{-1}}$.
  - **Profile:** The profile is broader (spanning the full 150 nm active layer) and non-uniform, showing a slope where gain is higher on one side (likely due to hole injection/mobility issues causing non-uniform carrier density).

2. Comparison and Comment

- **Magnitude Difference:** The Quantum Well achieves a peak material gain that is **over 20 times higher** than the Bulk laser $(1160\ VS\ 55\ cm^{-1})$.
- **Physical Interpretation:** This dramatic difference illustrates the primary advantage of quantum confinement. By restricting carrier motion to 2D, the density of states becomes step-like. This forces injected carriers to fill specific energy levels that contribute directly to gain at the lasing wavelength, rather than being spread out in energy and space as in the bulk material.
- **Efficiency:** The ability to generate such high gain $(> 1000\ cm^{-1})$ in a tiny volume allows the QW laser to overcome cavity losses with a much smaller total active volume, leading to significantly reduced threshold currents.

8. Spectral Characteristics

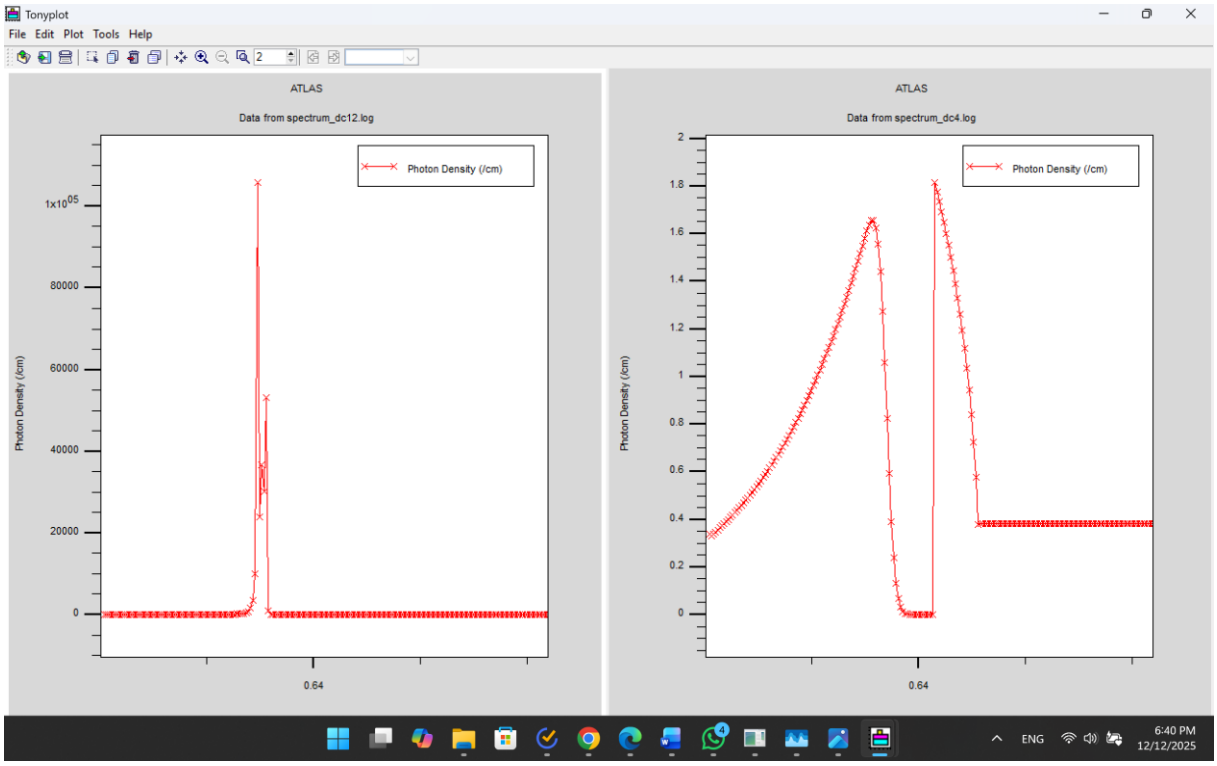


Figure [A5]: InGaP Quantum Well Comparison of Laser Photon Density versus Wavelength.

Right Panel: At $I = 1.5$ mA (Pre-Lasing). Left Panel: At $I = 5.5$ mA (Lasing).

Observations:

The simulation results clearly demonstrate the transition from spontaneous emission to stimulated emission as the injection current increases.

- **At $I = 1.5$ mA (Right Panel):**
  - **Regime:** The device is operating below the lasing threshold (LED Mode).
  - **Intensity:** The photon density is extremely low, peaking at approximately $\mathbf{1.8\ cm^{-3}}$.
  - **Spectral Width:** The spectrum exhibits a broad, This broadness corresponds to the thermal distribution of carriers within the energy bands, characteristic of incoherent **Spontaneous Emission**.
- **At $I = 5.5$ mA (Left Panel):**
  - **Regime:** The device is operating well above threshold (Laser Mode).
  - **Intensity:** The photon density has increased by five orders of magnitude, reaching a peak of $\mathbf{1 \times 10^5\ cm^{-3}}$.
  - **Spectral Width:** The spectrum has collapsed into a single, extremely sharp spike at the resonant wavelength.

Conclusion (Spectral Width Comparison):

Comparing the two graphs demonstrates the phenomenon of Spectral Narrowing. Below threshold, the output is broadband and incoherent. Above threshold, the high optical gain selectively amplifies the single cavity mode that satisfies the resonance condition, suppressing all other wavelengths. This results in a monochromatic (single frequency) output with a negligible spectral width, which is the defining characteristic of laser light.

Part B:

Deck Modifications:

```
# === MODIFIED REGION 1 (Cladding) ===
# Increased eg300 to 2.58 for better electrical confinement
# Decreased permittivity to 3 for better optical confinement
material region=1 eg300=2.58 NC300=9.13e+17 NV300=7.81e+18 \
         align=0.5 permittivity=3 epsinf=3 affinity=4.19 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10

material region=2 eg300=2.3 NC300=9.13e+17 NV300=7.81e+18 align=0.5 \
         permittivity=12.16 epsinf=12.16 affinity=4.26 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10

material region=3 taun0=1e-9 taup0=2e-8 copt=1.5e-10 epsinf=13.491 \
         align=0.5 eg300=1.92 affinity=4.4

material region=4 eg300=2.3 NC300=9.13e+17 NV300=7.81e+18 align=0.5 \
         permittivity=12.16 epsinf=12.16 affinity=4.26 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10

# === MODIFIED REGION 5 (Cladding) ===
# Matches Region 1 changes
material region=5 eg300=2.58 NC300=9.13e+17 NV300=7.81e+18 align=0.5 \
         permittivity=3 epsinf=3 affinity=4.19 \
         taun0=1e-9 taup0=2e-8 copt=1.5e-10
```

1. Energy Band Diagram of the SCS

Graph Reference:

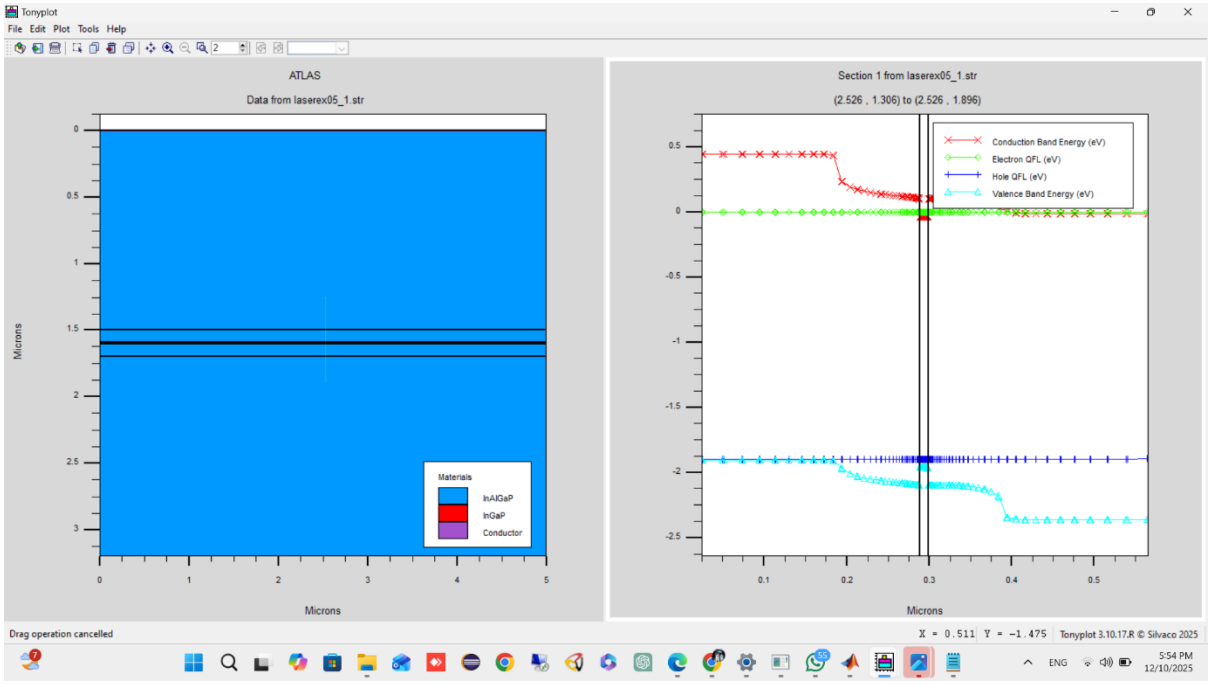


Figure [B1]: Y-Cut of the Energy Band Diagram for the QW Separate Confinement Structure (SCS).

Observations:

The energy band diagram) reveals a "staircase" potential profile characteristic of a Separate Confinement Heterostructure (SCH).

- **Cladding Layers (Regions 1 & 5):** The outermost layers exhibit the widest bandgap $(E_g = 2.58\ \text{eV})$, creating the highest potential barriers. This ensures strong electrical confinement, preventing carrier leakage.
- **Waveguide Layers (Regions 2 & 4):** The intermediate layers have a lower bandgap $(E_g = 2.2\ \text{eV})$, forming a "step" between the cladding and the active region.
- **Quantum Well (Region 3):** The central layer forms the deepest potential well $(E_g \approx 1.9\ \text{eV})$.

Comment:

This stepped structure provides a dual advantage:

1. **Carrier Funneling:** The intermediate energy steps help "funnel" carriers efficiently from the contact layers into the quantum well, increasing the probability of capture.
2. **Separate Confinement:** The electrical confinement is handled by the high-bandgap cladding, while the optical confinement (as we will see in the gain/intensity profiles) is handled by the refractive index contrast of the waveguide layers.

2. Recombination Rates Comparison

Graph Reference:

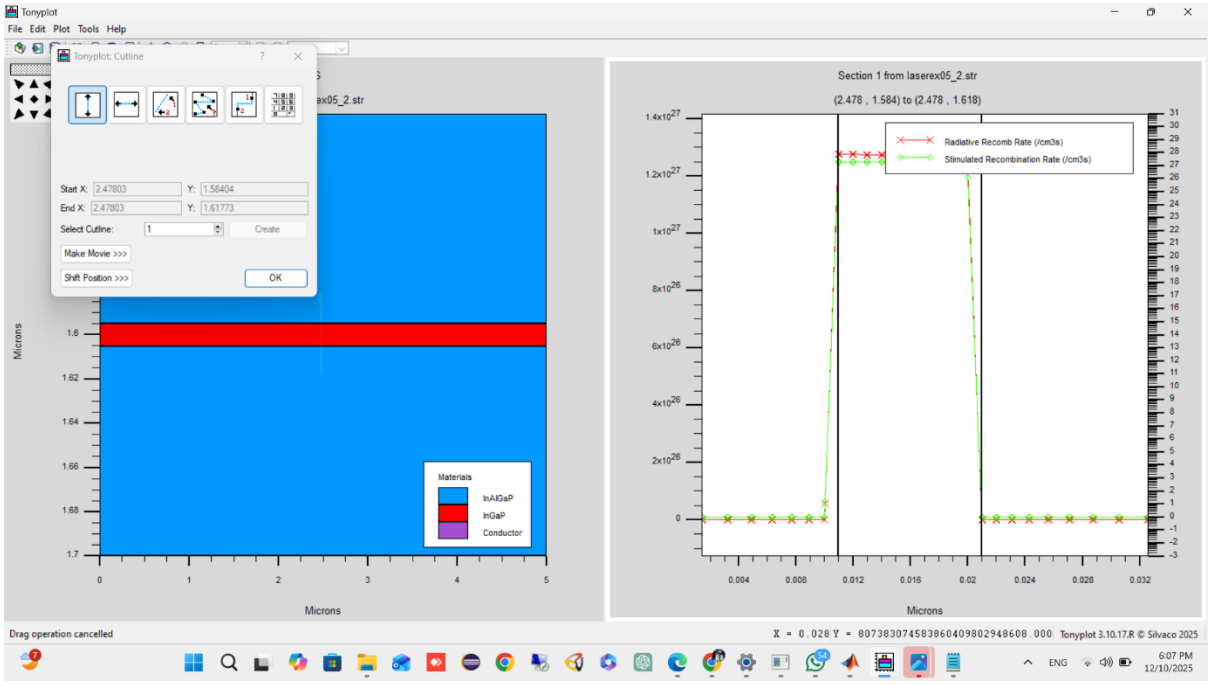


Figure [B2]: Stimulated and Radiative Recombination Rates in the SCS Active Region.

Observations:

- **Graph:** The graph shows the recombination rates along a vertical cut through the center of the device.
- **Confinement:** Just like in Part A, the recombination is strictly confined to the narrow **10 nm** Quantum Well region (between 0.015 μm and 0.025 μm on the zoomed X-axis).
- **Magnitudes:**
  - **Radiative Rate (Red):** Peaks at approximately $\mathbf{1.3 \times 10^{27}\ cm^{-3}s^{-1}}$.
  - **Stimulated Rate (Green):** Peaks at approximately $\mathbf{1.25 \times 10^{27}\ cm^{-3}s^{-1}}$.
  - The two rates are nearly identical, confirming that the device is lasing and the carrier density is clamped.

Comparison with Part A (Standard QW):

1. **Spatial Profile:** The profile shape is **identical** to Part A. This is expected because the Quantum Well width (10 nm) was not changed; only the surrounding cladding layers were modified.
2. **Magnitude:** The peak recombination rates are of the same order of magnitude ($\approx 10^{27}$).
3. **The SCS Effect:** While the *local* recombination physics inside the well remains largely the same, the **Separate Confinement Structure (SCS)** improves how efficiently the *generated photons* are used. By better confining the optical mode (via the refractive index step we created), the overlap between the photons and this recombination region is optimized, typically leading to a lower threshold current for the same output power.

3. Local Optical Gain Comparison

Graph Reference:

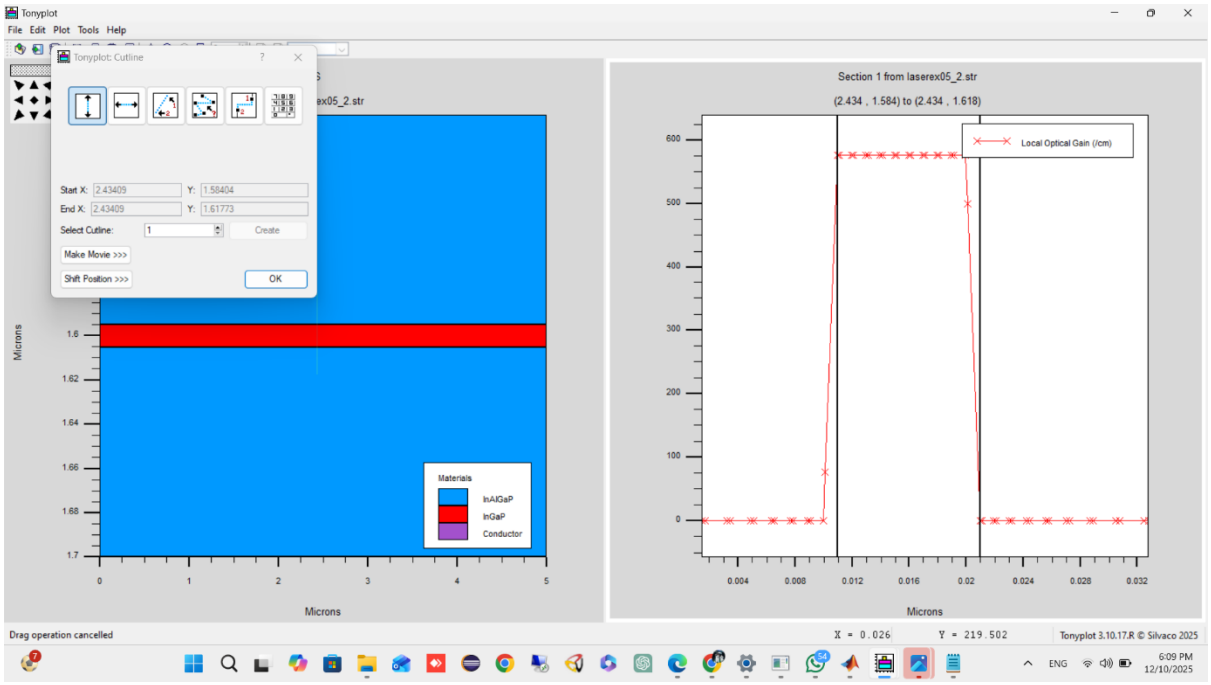


Figure [B3]: Local Optical Gain profile within the SCS Quantum Well.

Observations:

- **Profile:** Similar to Part A, the optical gain is strictly confined to the 10 nm Quantum Well layer.
- **Magnitude:** The peak local optical gain is approximately **580 cm^{-1}**.

Comparison with Part A (Standard QW):

- **Part A Peak Gain:** ≈ 1160 cm^{-1}.
- **Part B Peak Gain:** ≈ 580 cm^{-1}.
- **Result:** The required material gain has **decreased** by roughly half in the SCS structure.

At first glance, lower gain might seem "worse," but for a laser, this indicates superior performance.

- Threshold Condition: The condition for lasing is given by:

$$\Gamma \cdot g_{th} = \alpha_{total}$$

Where:

  - g_{th} is the material gain (what you plotted).
  - α_{total} is the total loss (mirrors + scattering).
  - Γ (Gamma) is the **Optical Confinement Factor** (how well the light overlaps with the active region).

- **The SCS Effect:** The modification in Part B (lowering permittivity in the cladding) created a strong waveguide, significantly increasing the **Optical Confinement Factor (Γ)**.
- **Conclusion:** Because Γ increased, the material gain (g_{th}) required to overcome the same losses **decreased**. It achieves lasing with less demand on the material because the light is guided more efficiently through the gain medium.

Part C:

Deck Modifications:

```
go atlas simflags="-p 20"

mesh space.mult=1.0 diag.flip width=300 auto
x.mesh loc=0.00 spac=0.50
x.mesh loc=1.50 spac=0.10
x.mesh loc=3.50 spac=0.10
x.mesh loc=5.00 spac=0.50
y.mesh loc=0.000 spac=0.2
y.mesh loc=1.500 spac=0.05
y.mesh loc=1.595 spac=0.001
y.mesh loc=1.605 spac=0.001
y.mesh loc=1.700 spac=0.05
y.mesh loc=3.200 spac=0.2

region num=1 material=InAlGaP y.min=0.0 y.max=1.5 x.comp=0.364 y.comp=0.156 \
     acceptor=5e18

region num=2 material=InAlGaP y.min=1.5 y.max=1.595 x.min=1.5 x.max=3.5 \
     x.comp=0.208 y.comp=0.312 acceptor=1e14

region num=3 material=InGaP y.min=1.595 y.max=1.605 x.min=1.5 x.max=3.5 \
     x.comp=0.516 donor=2e14

region num=4 material=InAlGaP y.min=1.605 y.max=1.7 x.min=1.5 x.max=3.5 \
     x.comp=0.208 y.comp=0.312 donor=1e14

region num=5 material=InAlGaP y.min=1.7 y.max=3.2 x.comp=0.364 y.comp=0.156 \
     donor=1e18

region num=6 material=InAlGaP y.min=1.5 y.max=1.7 x.max=1.5 \
     x.comp=0.364 y.comp=0.156 donor=2e18

region num=7 material=InAlGaP y.min=1.5 y.max=1.7 x.min=3.5 \
     x.comp=0.364 y.comp=0.156 donor=2e18

elec num=1 name=anode y.min=0.0 y.max=0.0
elec num=2 name=cathode y.min=3.2 y.max=3.2

material region=1 eg300=2.35 NC300=9.13e+17 NV300=7.81e+18 align=0.5 permittivity=11.39
material region=2 eg300=2.2  NC300=9.13e+17 NV300=7.81e+18 align=0.5 permittivity=12.16

material region=3 eg300=1.92 NC300=9.13e+17 NV300=7.81e+18 align=0.5 \
     permittivity=13.491 taun0=10e-9 taup0=50e-8 copt=1.5e-10

material region=4 eg300=2.2  NC300=9.13e+17 NV300=7.81e+18 align=0.5 permittivity=12.16
material region=5 eg300=2.35 NC300=9.13e+17 NV300=7.81e+18 align=0.5 permittivity=11.39
material region=6 eg300=2.35 NC300=9.13e+17 NV300=7.81e+18 align=0.5 permittivity=11.39
material region=7 eg300=2.35 NC300=9.13e+17 NV300=7.81e+18 align=0.5 permittivity=11.39

models fldmob srh optr auger fermi consrh
models region=3 yan ^optr spontaneous qwell well.ny=120 well.cnbs=1 well.vnbs=1
mobility mun=1000 mup=500 vsatn=7.7e6 vsatp=7.7e6

solve init
solve previous

output band.param con.band val.band u.srh u.radiat u.auger flowlines

probe name="nwell" nwell x=2.5 y=1.6
probe name="pwell" pwell x=2.5 y=1.6

solve vstep=0.05 name=anode vfinal=1.9

models fldmob srh optr auger fermi consrh gainmod=5 ^print

laser trap maxtrap=10 proj rf=8 rr=90 maxch=100 \
      photon.energy=1.8 einit=1.5 efinal=2.0 lmodes \
      neff=3.2 cavity.length=300 sin=1e-5 index.mode=1 ^coupled

solve previous

log outf=ProjectC.log

solve vstep=0.05 name=anode vfinal=1.95

save outfile=ProjectC_1.str

contact name=anode current
solve istep=0.5e-3 ifinal=1.0e-2 name=anode
save outfile=ProjectC_2.str

tonyplot ProjectC.log
quit
```

1. Total Light Output Power vs. Current

Graphs:

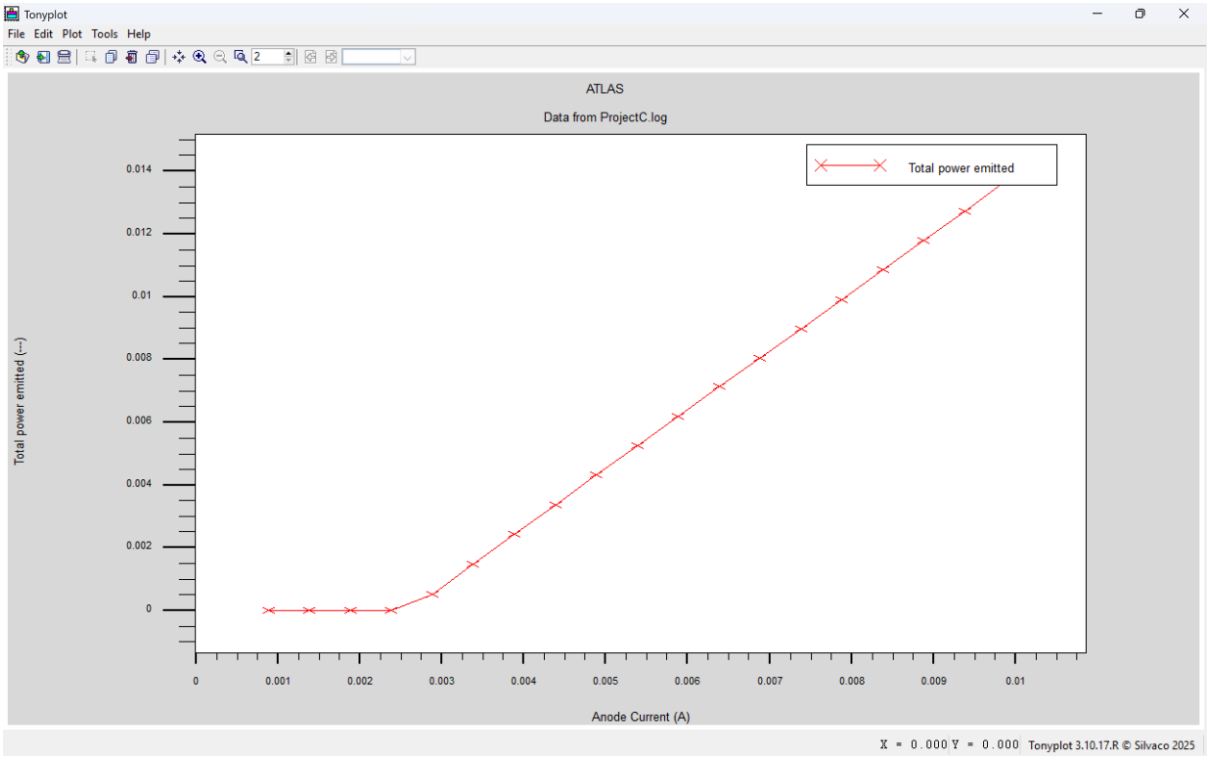


Figure [C1a]: Total Light Output Power versus Anode Current for the Buried Heterostructure (Part C).

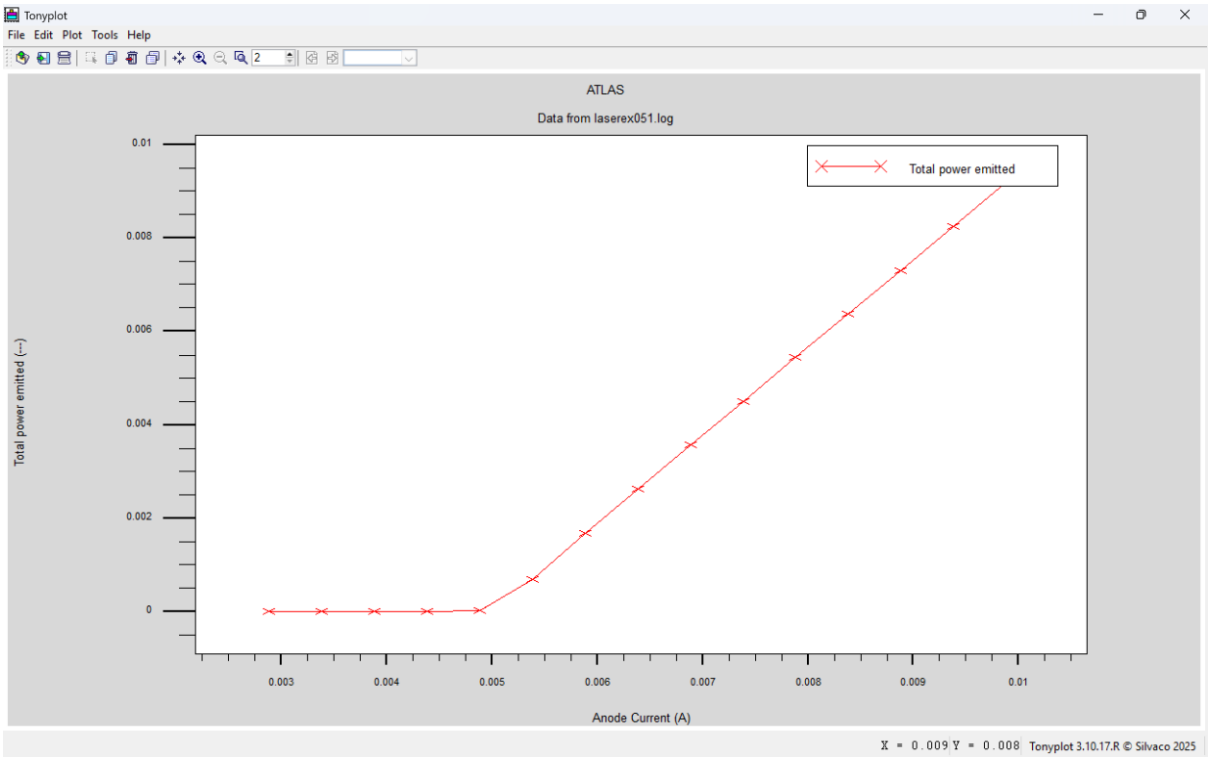


Figure [C1b]: Total Light Output Power versus Anode Current for the Stripe Geometry (Part A).

Observations:

- **Threshold Current (I_{th}):**
  - **Stripe Geometry (Part A):** The device begins lasing at approximately **5.0 mA**.
  - **Buried Heterostructure (Part C):** The device begins lasing at approximately **3 mA**.
- **Comparison:** The Buried Heterostructure reduces the threshold current by **40%** compared to the Stripe Geometry structure.

Comment and Analysis:

The significant reduction in threshold current in the Buried Heterostructure (BH) is driven by the simultaneous improvement of both Carrier Confinement and Optical Confinement:

1. **Carrier Confinement (Electrical):**
   - **Stripe Geometry:** In Part A, current is injected through a top contact but lacks lateral barriers. This allows carriers to **spread laterally** (current spreading) into regions where there is no optical mode. This "leakage" means a higher total current is needed to reach the required carrier density for inversion.
   - **Buried Heterostructure:** In Part C, the active region is surrounded by burying layers with a larger bandgap. This creates a potential barrier that strictly confines the carriers to the active region, eliminating lateral leakage. This forces the injected current to contribute more efficiently to gain, lowering the threshold.
2. **Optical Confinement (Waveguiding):**
   - **Stripe Geometry:** Part A structure relies on **Gain Guiding**, where the optical mode is loosely confined only by the presence of gain. This results in a poor overlap between the optical mode and the gain region (low Confinement Factor, Γ).
   - **Buried Heterostructure:** The BH structure relies on **Index Guiding**. The burying layers have a lower refractive index than the active region, creating a physical waveguide that traps the light tightly in both vertical and lateral directions. This increases the Confinement Factor (Γ). Since the threshold gain is inversely proportional to confinement ($g_{th} \propto 1/\Gamma$), the better overlap further reduces the gain required to lase.

Conclusion: By preventing current from leaking out and preventing light from spreading out, the Buried Heterostructure achieves lasing with significantly less input power.

2. Light Intensity Distribution & Mode Analysis

Graphs:

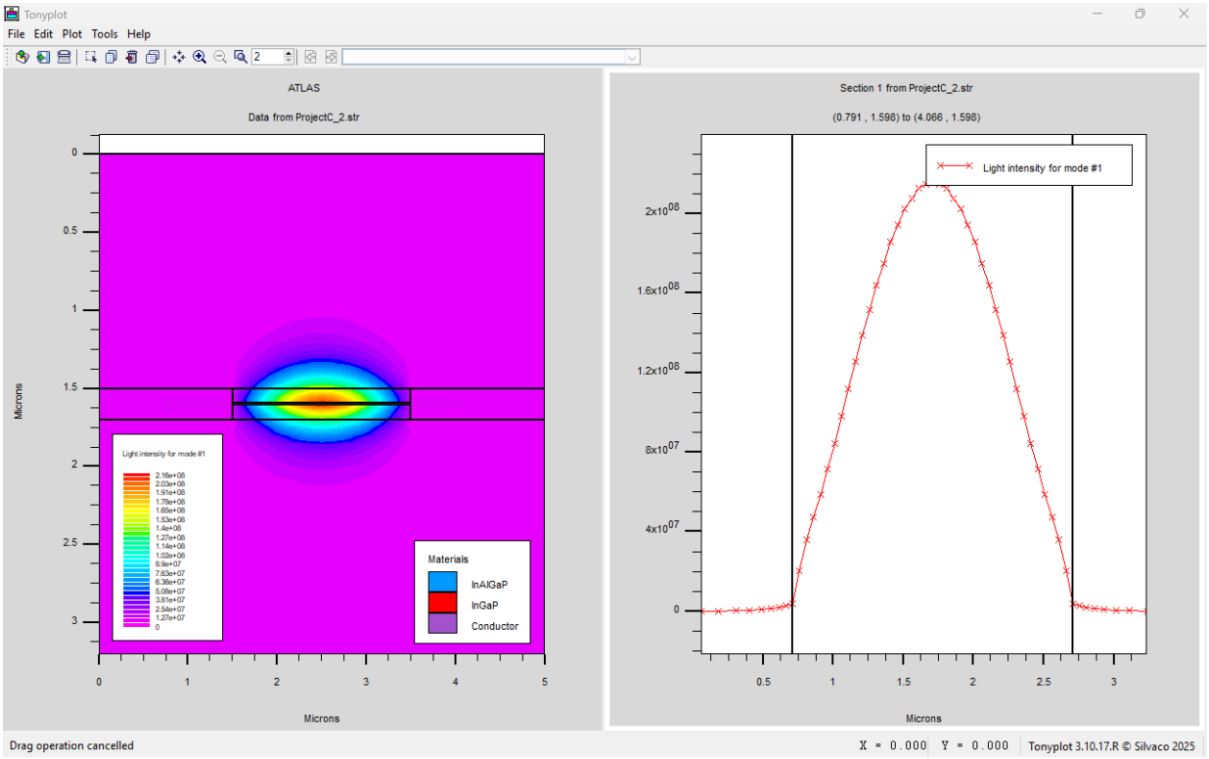


Figure [C2a]: 1D Horizontal Light Intensity Profile for the Buried Heterostructure (Part C).

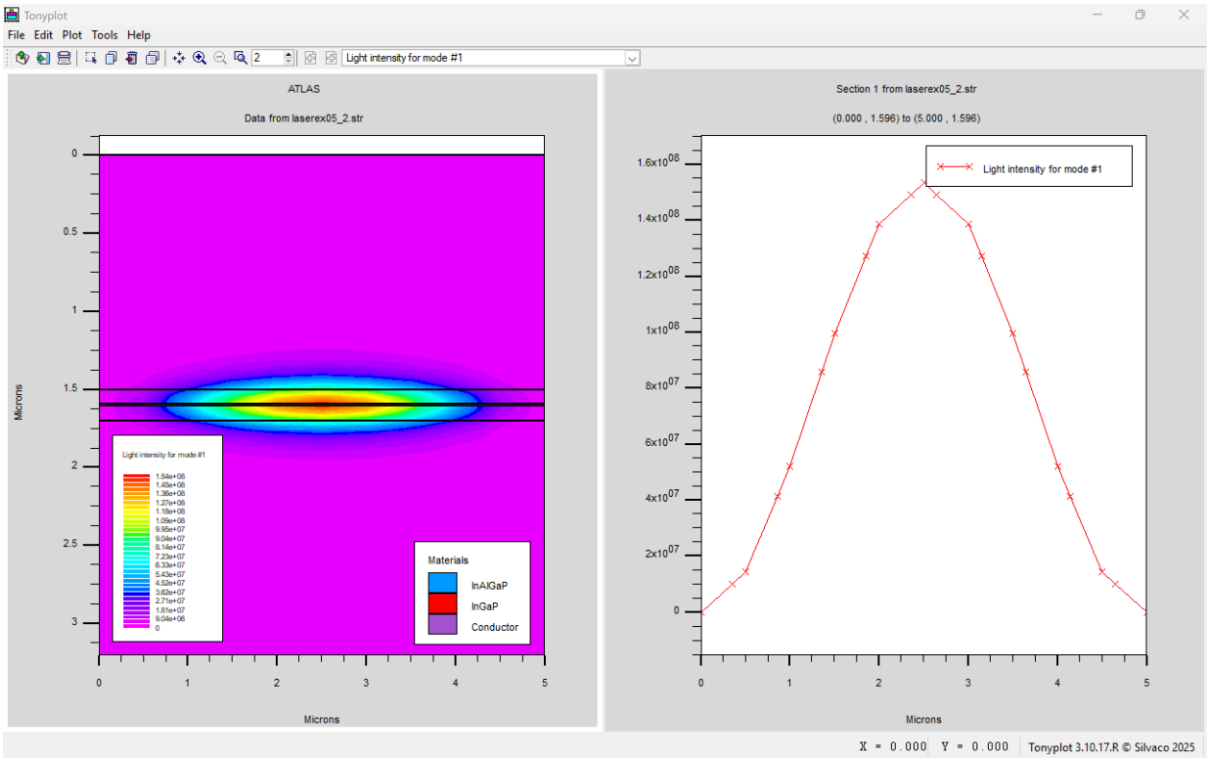


Figure [C2b]: 1D Horizontal Light Intensity Profile for the Stripe Geometry (Part A).

Observations:

1. **Stripe Geometry (Figure A):** The horizontal intensity profile is broad and relatively flat. The optical mode tails decay slowly, indicating that light is spreading laterally into the surrounding regions. There are no sharp boundaries confining the light.
2. **Buried Heterostructure (Figure B):** The horizontal intensity profile is significantly sharper and narrower. The intensity forms a perfect Gaussian shape that drops to zero precisely at the boundaries of the active region ($x = 1.5\ \mu m$ and $x = 3.5\ \mu m$).

Comment and Analysis:

- Comparison of Optical Confinement Mechanisms:

The difference in these horizontal cuts illustrates the transition from Gain Guiding to Index Guiding.

  - **Stripe Geometry (Gain Guiding):** In Part A, the refractive index is uniform laterally. The light is confined only because the gain is highest in the center (where current is injected). As seen in Figure A, this weak confinement allows the optical mode to "diffract" or spread laterally into the unpumped, lossy regions, leading to a wider spot size and lower efficiency.
  - **Buried Heterostructure (Index Guiding):** In Part C, the active region is physically bounded by burial layers with a lower refractive index. This creates a built-in waveguide (a refractive index step) that strictly reflects light back into the core. Figure B confirms this: the light is physically prevented from spreading beyond the $2.0\ \mu m$ active width.

- Transverse Mode Verification (Deliverable #3):

The Buried Heterostructure is operating in a Single Fundamental Transverse Mode (TE_{00}).

  - **Evidence:** The intensity profile in Figure B shows exactly one distinct peak with no side lobes.
  - **Reason:** By reducing the active region width to $\mathbf{2.0\ \mu m}$ and burying it, the waveguide becomes too narrow to support higher-order modes. This acts as a spatial filter, forcing the laser to oscillate purely in the fundamental mode, which is ideal for coupling into optical fibers.

3. Spectral Characteristics (Additional Confirmation)

Graph:

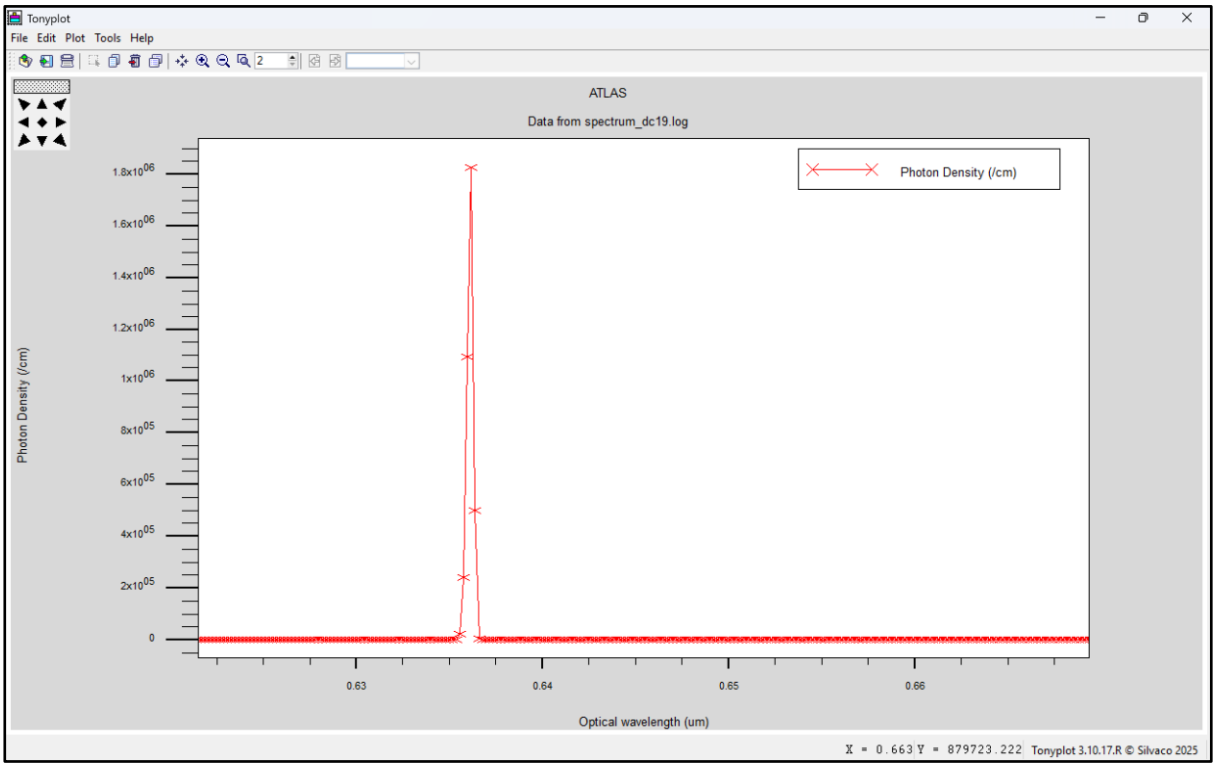


Figure [C3a]: Laser Photon Density vs. Optical Wavelength at $I = 10$ mA for the Buried Heterostructure.

Observations:

- The spectrum exhibits a single, extremely sharp peak at approximately **636nm**.
- The spectral width is negligible, and there are no visible side modes or secondary peaks in the plotted range.

Comment and Analysis:

- Spectral Purity:

While the intensity profiles confirmed that the laser operates in a single transverse mode (spatial confinement), this spectral plot confirms that the laser also operates in a single longitudinal mode.

- Correlation:

The strong optical confinement provided by the Buried Heterostructure not only stabilizes the spatial beam but also improves the spectral purity. The high gain density in the narrow active region allows one specific cavity mode to dominate completely, suppressing all other wavelengths. This monochromatic output is a signature of a well-designed index-guided laser.

Part D:

Deck Modifications:

```
go atlas
mesh  space.mult=1.0 diag.flip
#
x.mesh loc=0.00 spac=0.5
x.mesh loc=5 spac=0.5

y.mesh loc=0.00 spac=0.15
y.mesh loc=1.5 spac=0.01
y.mesh loc=1.6 spac=0.001
y.mesh loc=1.61 spac=0.01
y.mesh loc=1.71 spac=0.15
y.mesh loc=3.21 spac=0.15

region num=1 material=InP y.max=1.6


region  num=2  material=InGaAsP  y.min=1.6 y.max=1.61 x.comp=0.25 y.comp=0.5

region num=3 material=InP y.min=1.61

elec     num=1  name=cathode    bottom
#
elec     num=2  name=anode      top
#
doping     uniform reg=1  p.type conc=1.e18
doping     uniform reg=2  n.type conc=2.e14
doping     uniform reg=3  n.type conc=1.e18

material material=InP     taun0=2.e-9  taup0=2.e-9  copt=1.5e-10 \
 mun=2400.0 mup=80.0 align=0.6
#
material material=InGaAsP taun0=10.e-9 taup0=10.e-9 copt=1.5e-10 \
 mun=4600.0 mup=150.0 gainlmin=2.5e-16
#
models
models material=InP     fldmob  srh optr fermi print
models material=InGaAsP fldmob  srh optr fermi print
#
solve init

output con.band val.band
save outf=laserex01_0.str

method  newton autonr trap
solve v2=0.01
solve v2=0.05
solve v2=0.1
solve v2=0.2
solve v2=0.4
solve v2=0.6
#
probe name="nwell" nwell x=2.5 y=1.605
probe name="pwell" pwell x=2.5 y=1.605

models material=InGaAsP fldmob srh optr fermi print laser gainmod=2 las_maxch=50

laser maxch=50 photon.energy=1.025 cavity.length=50 \
      einit=0.9 efinal=1.3  lmodes neff=3.2

log  outf=laserex01_1.log
#
solve v2=0.8
save outfile=laserx01_3.str
solve v2=0.9
solve v2=1.0
solve v2=1.1

output con.band val.band recomb u.srh u.aug u.rad flowlines
solve vstep=0.05 electr=2 vfinal=1.7

save outfile=laserex01_1.str

quit
```

1. Quantum Well Thickness

Same thickness as part A , 10nm

2. Main Changes in Models and Bias

To simulate the Quantum Well device (Part D) compared to the Bulk device (Assignment 2 / Example 1), three critical changes were made to the physics models and simulation structure:

1. **Gain Model (gainmod):**
    - **Assignment 2 (Bulk):** Used gainmod=2. This model calculates optical gain based on a parabolic (3D) density of states suitable for bulk crystals.
    - **Part D (QW):** Used gainmod=5. This model is specifically designed for Quantum Wells. It calculates gain based on the "step-like" (2D) density of states and accounts for the quantization of energy levels in the well.
2. **Quantum Capture (qwell):**
    - **Assignment 2:** Used standard bulk recombination models.
    - **Part D:** Added the models region=2 ... qwell statement. This activates the Schrödinger-Poisson solver for the active region, calculating the confined sub-band energy levels and wavefunctions necessary for QW physics.
3. **Mesh Resolution:**
    - **Assignment 2:** Used a coarse mesh ($0.02\mu m$)for the active region.
    - **Part D:** Required a hyper-fine mesh ($0.001\mu m/1nm$) in the active layer to accurately resolve the quantum wavefunctions.

3. Energy Band Diagram Analysis

Graph:

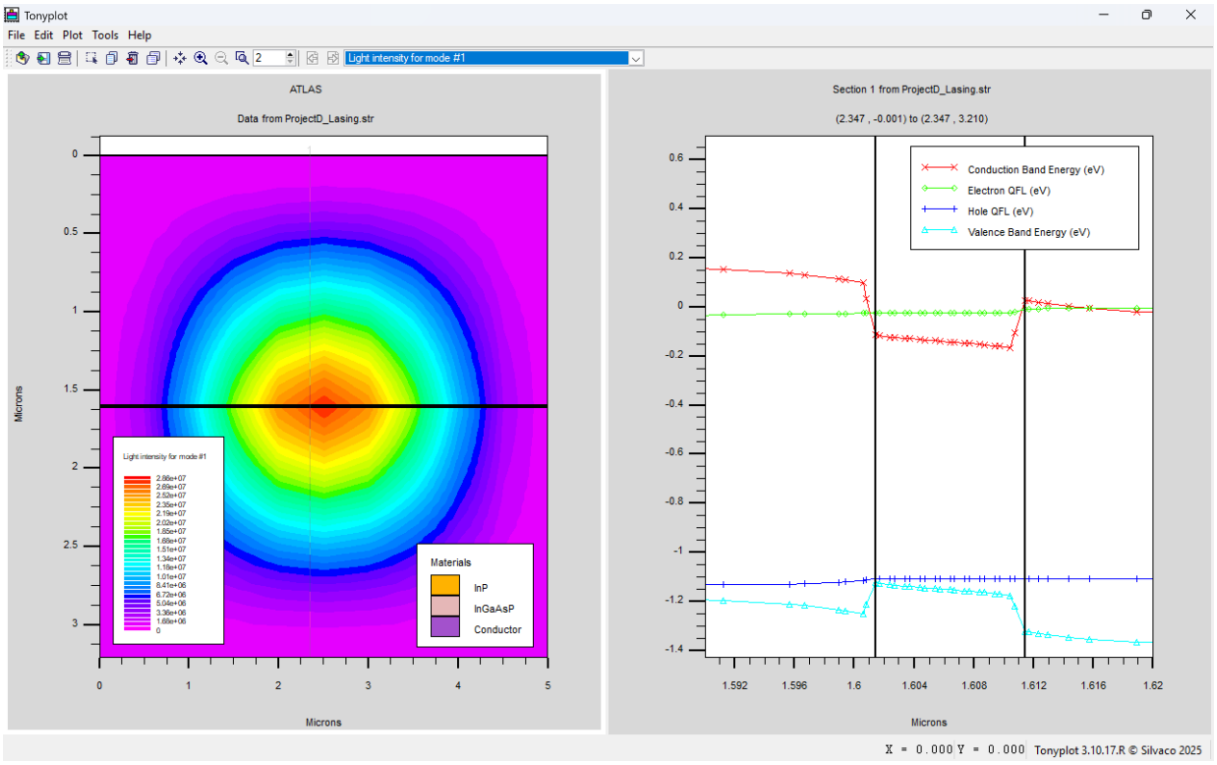


Figure [D1]: Vertical Energy Band Diagram for the Standard InP/InGaAsP SQW Laser (Part D).

Observations:

1. **Structure Profile:**
   - The diagram displays a classic **Type-I Single Quantum Well** profile.
   - **Barriers (Cladding):** The outer regions (Left and Right) are **InP**, which has a wide bandgap $(E_g \approx 1.35\ \text{eV})$.
   - **Active Region (Well):** The center shows a single, sharp, rectangular "notch" of **InGaAsP** $(E_g \approx 0.95\ \text{eV})$.
2. **Geometric Confinement:**
   - Unlike the "Staircase" design, this structure transitions directly from Cladding to Well.
   - The width of the central notch corresponds exactly to the **10 nm** $(0.01\mu m)$ thickness defined in the code (y.min=1.60 to y.max=1.61).

- **Part D (Standard SQW):** The active region is reduced to a **thin quantum line** (10 nm).
    - **Carrier Confinement:** Both structures trap carriers, but the SQW does so in a volume that is **15 times smaller**. This creates a 2D "step-like" density of states, forcing electrons and holes to populate specific energy sub-bands rather than a continuous 3D distribution.
    - **Optical Confinement Difference:** In the Bulk laser, the 150 nm active layer was thick enough to help guide the light. In this Standard SQW, the 10 nm well is too thin to effectively guide the optical mode on its own. Without the SCH layers (the "steps"), the optical mode in this design will spread significantly into the InP cladding, leading to a lower Optical Confinement Factor (Γ) compared to the Bulk device.

4. Checking for Population Inversion

Population inversion is the necessary condition for lasing, where the probability of finding an electron in the conduction band is higher than in the valence band.

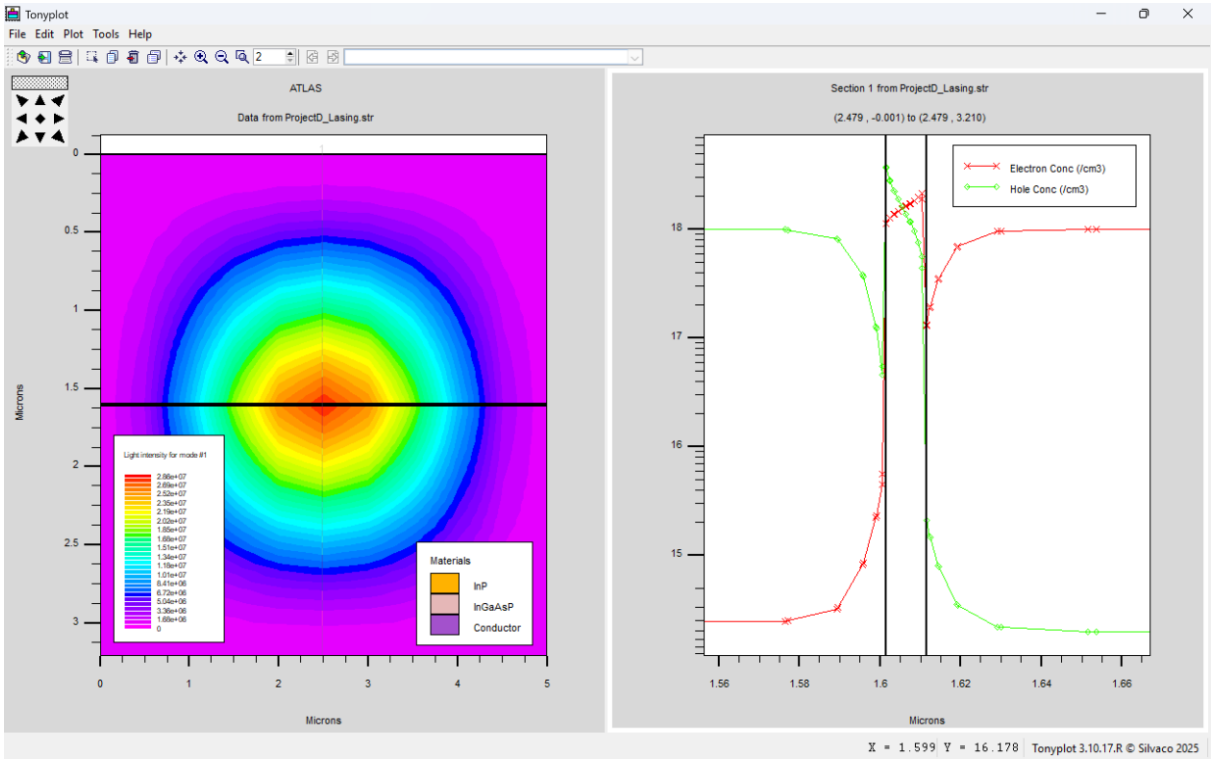
- **Method:** We inspect the **Quasi-Fermi Levels (QFL)** in the band diagram.
- **Observation:** Inside the 10 nm well, the **Electron QFL** moves *into* the Conduction Band, and the **Hole QFL** moves *into* the Valence Band.
- Condition: The separation between the quasi-Fermi levels is greater than the bandgap energy:

$$F_n - F_p > E_g$$

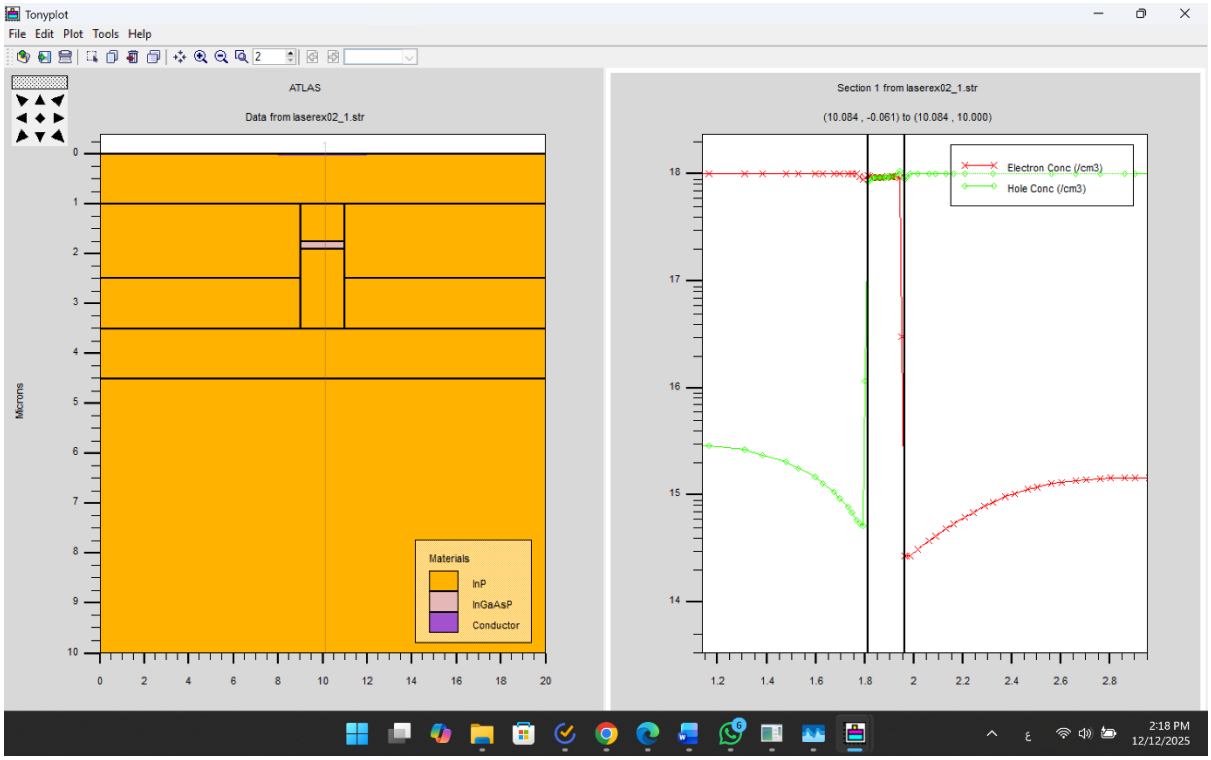
This "crossing" of the Fermi levels into the bands confirms that the active region is degenerate and population inversion has occurred.

5. Electron Concentration in the Active Region

Graphs Analyzed:



- **Figure [D2a] (SQW Part D):** Electron Concentration Vertical Cut.



- **Figure [D2b] (Bulk Assignment 2):** Electron Concentration Vertical Cut.

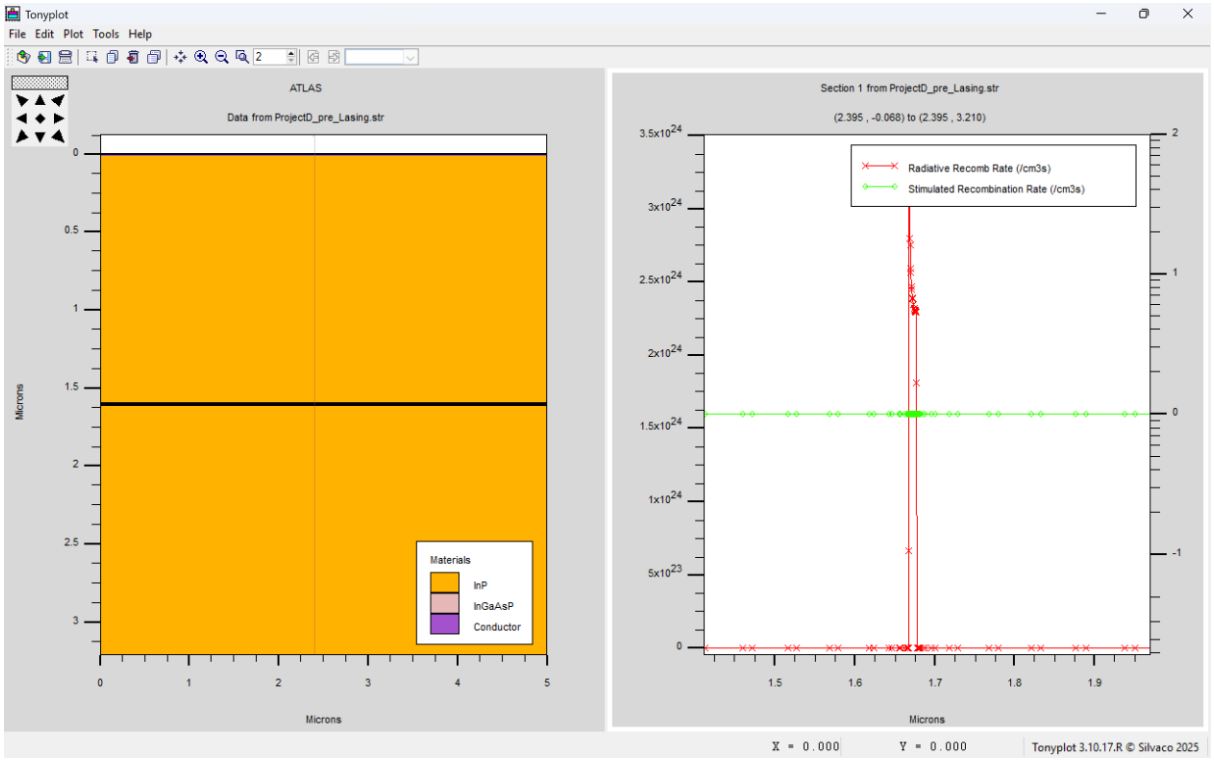
Observations & Comparison:

1. **Peak Magnitude:**
    - **SQW (Part D):** The electron concentration peaks inside the well at approximately $\mathbf{2 - 3 \times 10^{18}\ cm^{-3}}$.
    - **Bulk (Assignment 2):** The electron concentration plateaus across the active region at a slightly lower level, around $\mathbf{1 - 2 \times 10^{18}\ cm^{-3}}$.
    - **Result:** The Quantum Well achieves a slightly higher peak carrier density, but the values are in the same order of magnitude.

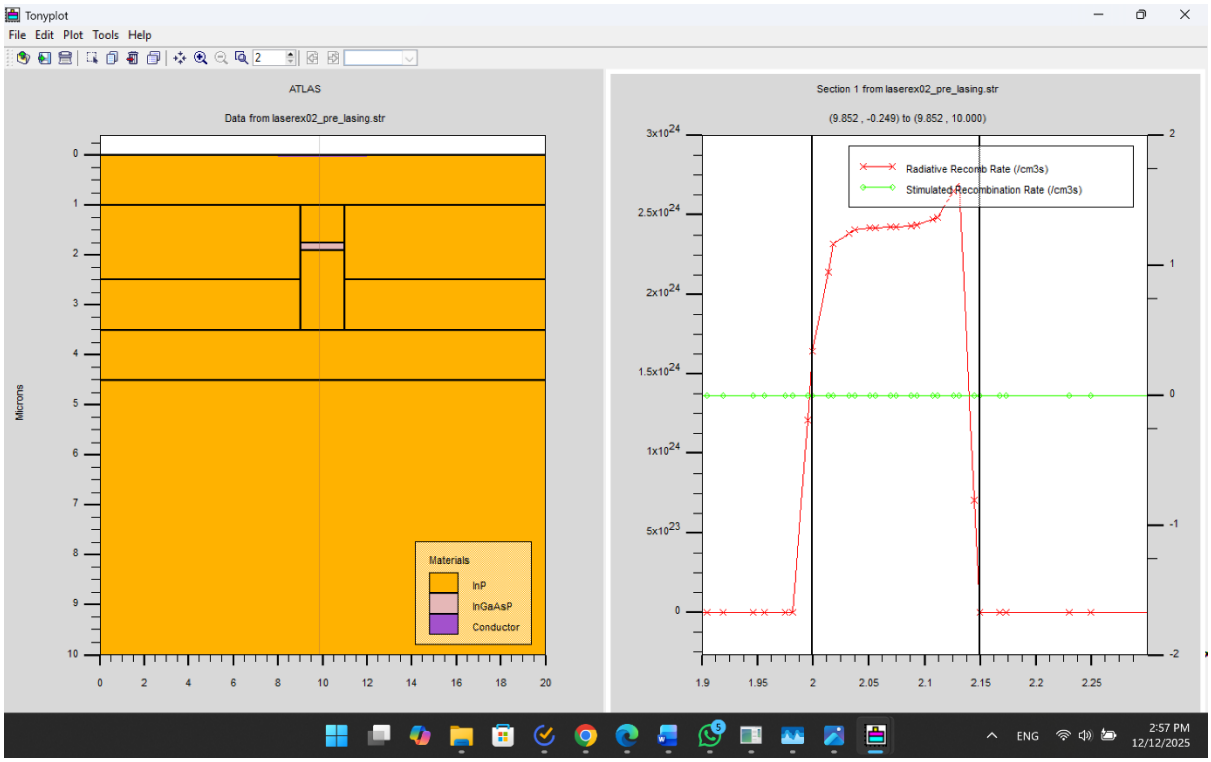
6. Recombination Rates Analysis

In this section, we analyze the transition from spontaneous emission (LED mode) to stimulated emission (Lasing mode) in the Standard SQW laser and compare it to the Bulk laser from Assignment 2.

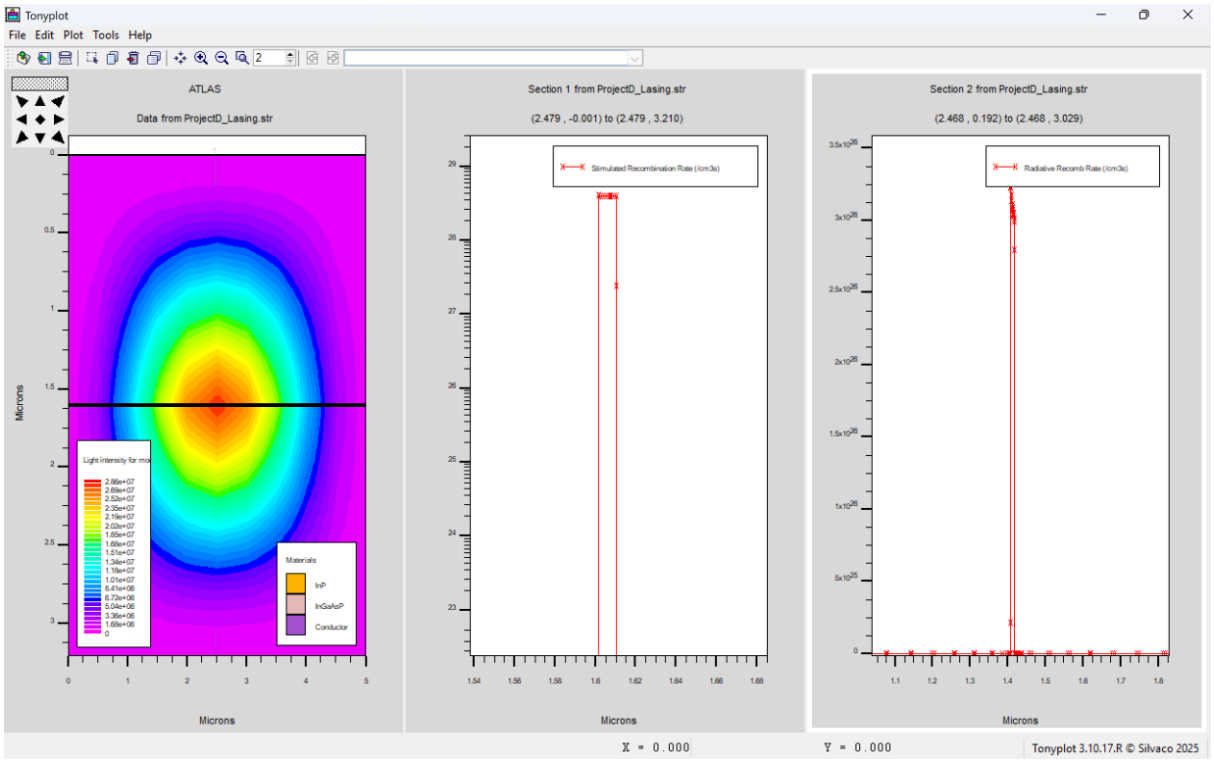
Graphs Analyzed:



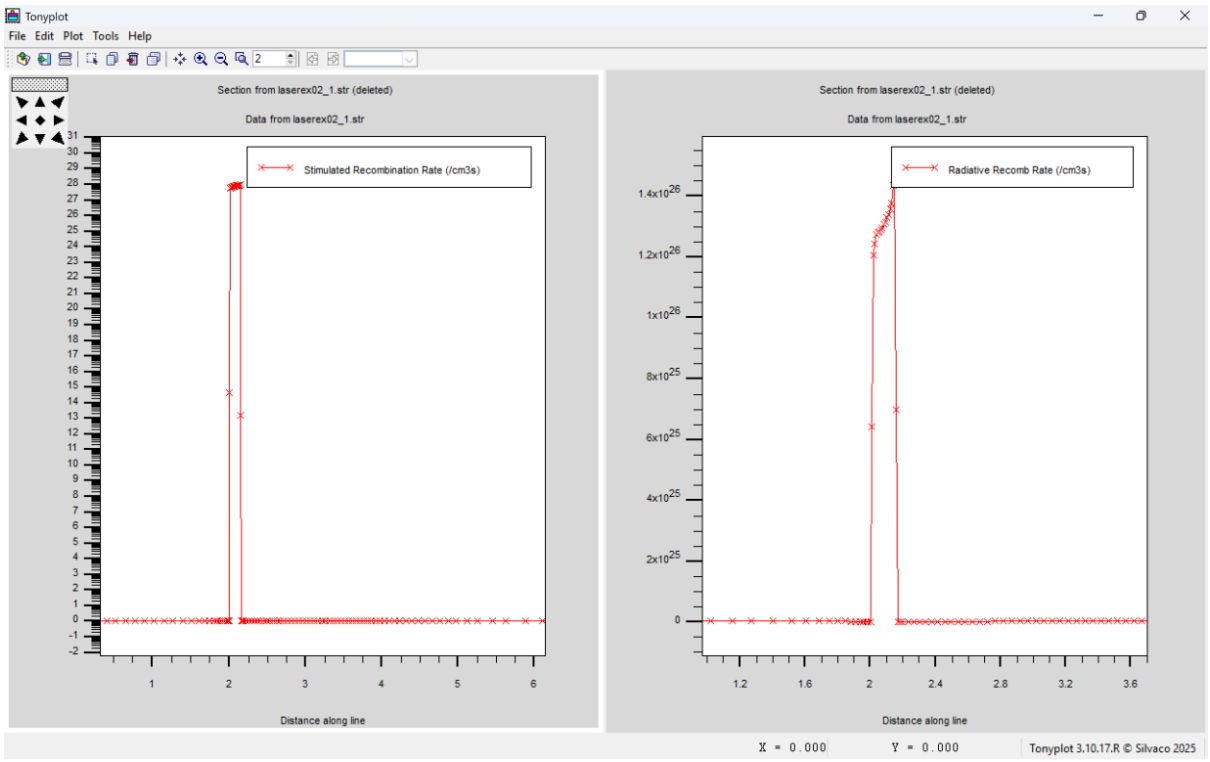
- **Figure [D3a]: SQW Pre-Lasing Rates.**



- **Figure [D3b]: Bulk Pre-Lasing Rates.**



- **Figure [D3c]: SQW Post-Lasing Rates.**



- **Figure [D3d]: Bulk Post-Lasing Rates.**

A. Pre-Lasing Behavior (LED Mode)

Observation:

Below the threshold current, the device operates as a Light Emitting Diode (LED).

- **Radiative Recombination (Red): Dominates the process. In the SQW (Figure D6a), this rate peaks at approximately 3×10^{24} $cm^{-3s^{-1}}$ and is strictly confined to the 10 nm quantum well.**
- **Stimulated Recombination (Green): Is effectively zero (flat line), indicating that the photon density inside the cavity is not yet high enough to trigger coherent amplification.**

B. Post-Lasing Behavior (Laser Mode)

Observation:

Above the threshold current, the device switches to lasing operation.

- **Stimulated Recombination (Green): Spikes dramatically, reaching a massive peak of approximately 3×10^{28} $cm^{-3s^{-1}}$. This rate is two orders of magnitude higher than the radiative rate (10^{26}), confirming that stimulated emission is now consuming nearly all injected carriers.**
- **Clamping: The radiative rate stops increasing (clamps) because the carrier density is pinned at the threshold value.**

C. Comparison with Assignment 2 (Bulk Laser)

1. Peak Magnitude Similarity:

Remarkably, the local peak reaction rates are similar in both devices.

- **Pre-Lasing: Both devices show radiative peaks in the 10^{24} range.**
- **Post-Lasing: Both devices achieve stimulated emission peaks in the 10^{28} range.**

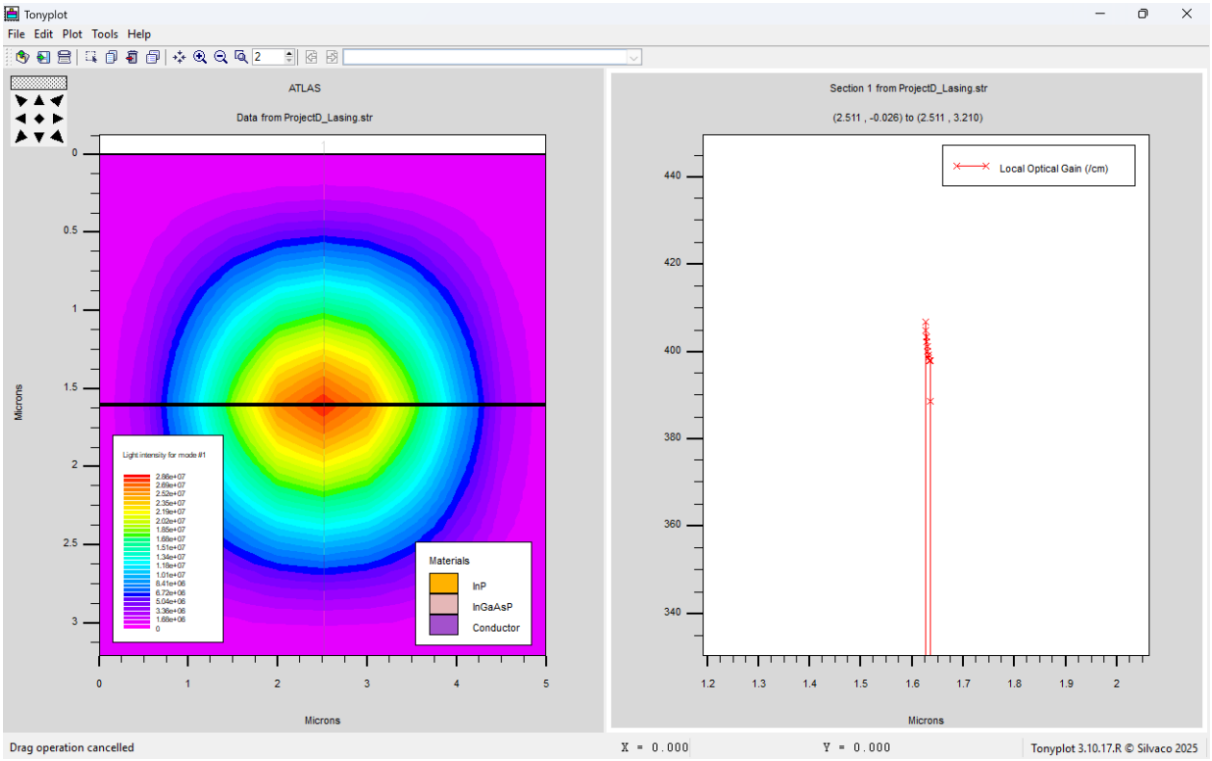
2. The Geometric Difference:

The critical difference lies in the Volume where this reaction occurs.

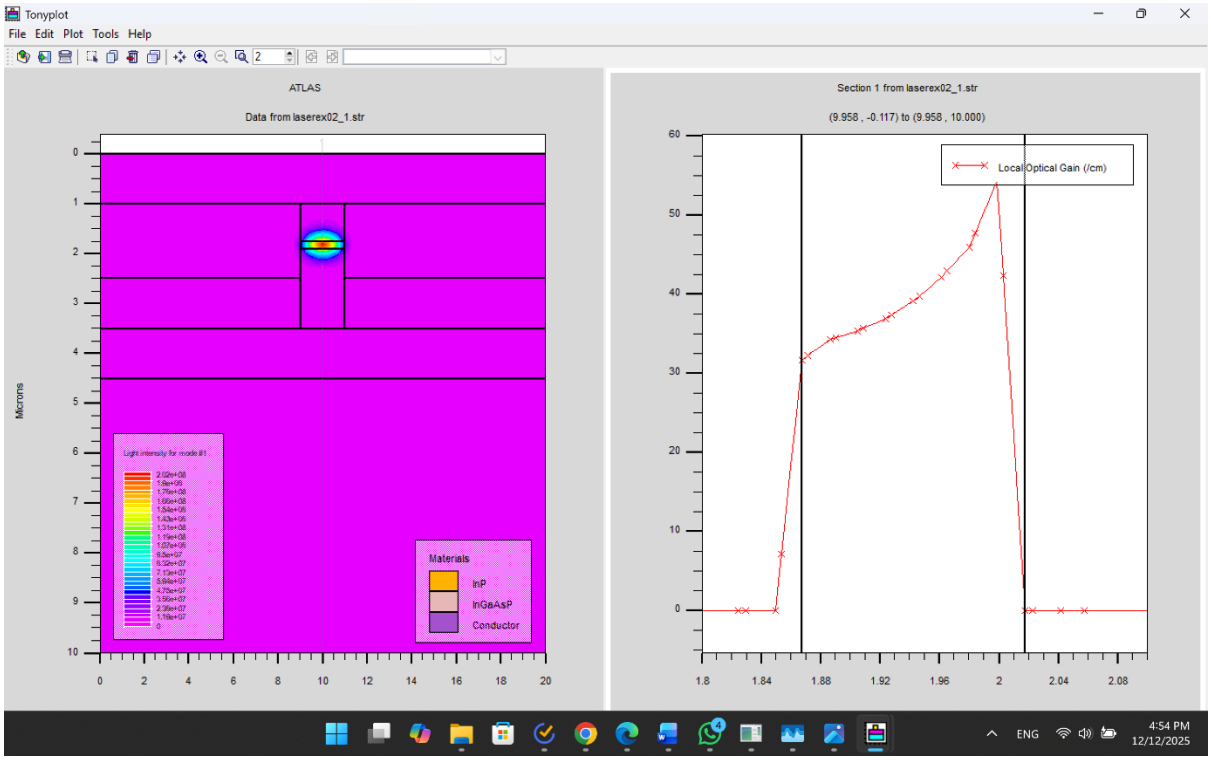
- **Bulk Laser: The high recombination rate spans a thick 150 nm slab.**
- **SQW Laser: The same intense recombination rate is concentrated into an ultra-thin 10 nm sheet.**

7. Local Optical Gain Analysis

Graphs Analyzed:



- **Figure [D4a] (SQW Part D):** Local Optical Gain Vertical Cut.



- **Figure [D4b] (Bulk Assignment 2):** Local Optical Gain Vertical Cut.

Observations :

- **Peak Value:** The Local Optical Gain spikes to approximately **410 cm^{-1}**.
- **Profile:** The gain profile is extremely sharp, resembling a delta function, and is strictly confined to the **10 nm** width of the quantum well.
- **Context:** This high value indicates that the material within the well is being pumped very hard to generate sufficient amplification.

Comparison with Assignment 2 (Bulk):

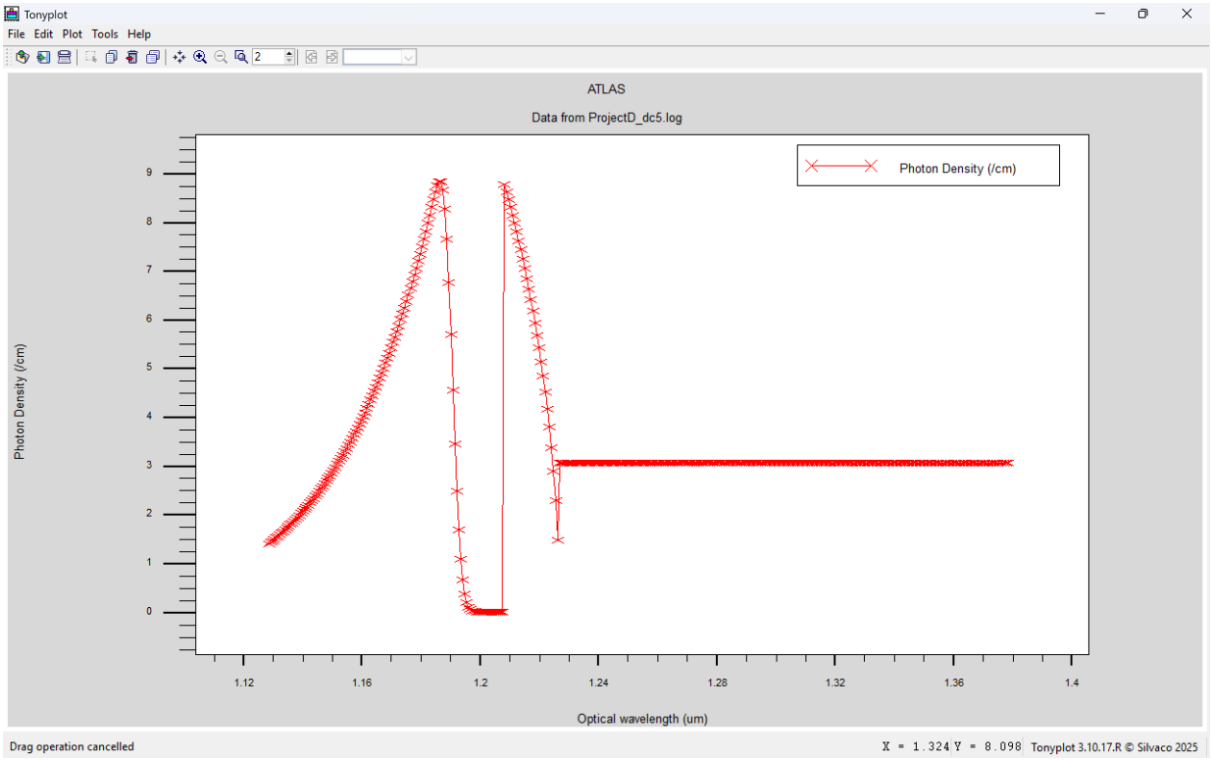
- **Magnitude Difference:**
  - **Bulk Laser:** The gain peaks at a much lower value of approximately **54 cm^{-1}**.
  - **SQW Laser:** The gain is roughly **7.5 times higher** (410 vs 54).
- **The "Gain-Confinement" Trade-off:**
  - **Bulk:** The active region is thick (150 nm), so the Optical Confinement Factor (Γ) is high (the light overlaps well with the gain medium). Because Γ is high, the required material gain (g_{th}) can be low.
  - **SQW:** The active region is tiny (10 nm), so the Optical Confinement Factor (Γ) is very low (most light travels in the cladding). To compensate for this poor overlap and still satisfy the lasing condition ($\Gamma \cdot g_{th} = \text{Loss}$), the material itself must generate significantly higher local gain.

Conclusion:

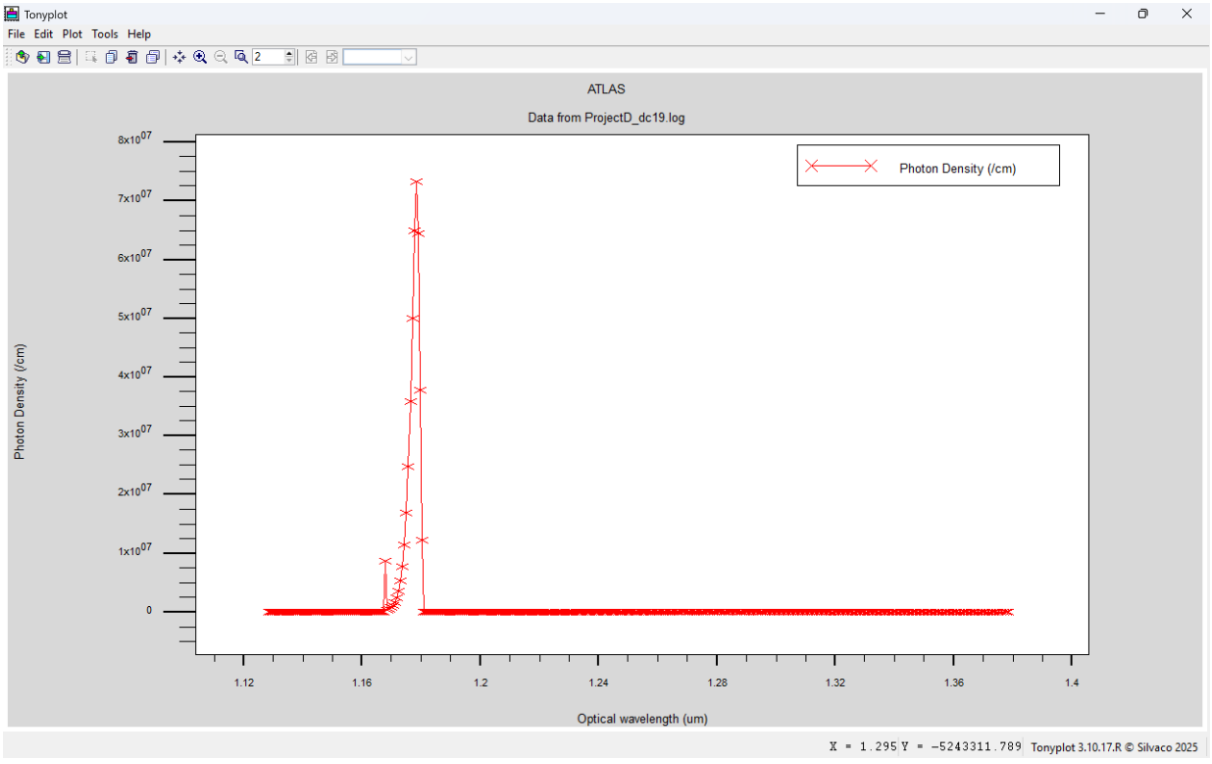
The comparison proves that the Quantum Well is a much more potent gain medium. It generates nearly an order of magnitude more gain per unit length (410 cm^{-1}) than the bulk material (54 cm^{-1}), allowing the device to lase despite having only a fraction of the active volume.

8. Lasing Spectrum and Spectral Width Analysis

Graphs Analyzed:



- Figure [D5a] (Pre-Lasing Spectrum): Photon Density vs. Wavelength at low current.



- Figure [D5b] (Post-Lasing Spectrum): Photon Density vs. Wavelength at high current.

Observations:

1. Pre-Lasing Behavior (Figure D8a):

- **Profile:** The spectrum shows a broad, "double-hump" profile with emission spread widely between 1.16μm and 1.22μm.
- **Spectral Width:** is broad, roughly $\mathbf{0.04 - 0.05\mu m}$ $(40 - 50$ nm$)$.
- **Physics:** This broad output is characteristic of **Spontaneous Emission (LED mode)**. Incoherent photons are being emitted over a wide range of energies corresponding to the thermal distribution of carriers in the bands.

2. Post-Lasing Behavior (Figure D8b):

- **Profile:** The spectrum collapses into a single, dominant, ultra-sharp spike centered precisely at $\mathbf{1.17\mu m}$.
- **Spectral Width:** The width of this peak is extremely narrow, appearing to be less than $\mathbf{0.002\mu m}$ $(< 2$ nm$)$.
- **Physics:** This dramatic **Spectral Narrowing** is the definitive proof of **Stimulated Emission**. The optical cavity has selected the longitudinal mode at 1.17μm, and stimulated emission has amplified this specific wavelength while suppressing the side modes and the broadband background.

Conclusion:

The transition from a broad 50 nm hump to a sharp < 2 nm spike at 1.17μm confirms that the Standard SQW laser is operating correctly in coherent single-mode.